Bright Minds, Big Rent:
Gentrification and the Rising Returns to Skill

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This paper is a revised version of *Bright Minds, Big Rent: Gentrification and the Rising Returns to Skill. (CES 16-36R) from August, 2016. A copy of the original paper is available upon request.*

CES 16-36R November, 2016

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Abstract

In 1980, Census data indicate, housing prices in large US cities rose with distance from the city center. By 2010, the relationship had reversed. We propose that this development can be traced to high-income households working longer hours. With little non-market time, proximity to work takes on added salience, leading high-income households to forgo suburban amenities and extending the gentrification trend beyond its 1970s niche status. In a tract-level data set covering the 27 largest US cities, years 1980-2010, we find support for our hypothesis. Using a Bartik-type demand shifter for skilled labor we find that full-time skilled workers favor centrality and the rising share in the population can account for the observed price changes in favor of the city center.

Keyword*:* Gentrification; suburbanization; returns to skill; labor supply; location choice

JEL Classification: J13, J24

[*] We have benefitted from discussions with Joe Altonji, Nathan Baum-Snow, V. V. Chari, Julie Cullen, David Deming, Jonathan Fisher, Jonathan Heathcote, Matthew Kahn, Jeanne Lafortune, Shirley Liu, Douglas Lucius, Chris Mayer, Enrico Moretti, Marcelo Moreira, Jordan Rappaport, Bernard Salanie, Aloysius Siow, seminar participants at the Minneapolis Federal Reserve, the Barcelona Graduate Summer School 2015, The Econometric Society World Congress August 2015, 2016 Research Symposium on Gentrification and Neighborhood Change at the Federal Reserve Bank of Philadelphia, 2016 Canadian Women Economists Network Luncheon, 2016 mini conference on family/labor/development economics at CUHK, Simon Fraser University. We thank the New York Federal Statistical Research Data Centers, Baruch, Center for Economic Studies, U.S. Census Bureau. Any opinions and conclusions expressed herein are those of the authors and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed.

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I know things will get better
You'll find work and I'll get promoted
We'll move out of the shelter
Buy a bigger house and live in the suburbs

Tracy Chapman, Fast Car, 1988

The truth is that we are living at a moment in which the massive outward migration of the affluent that characterized the second half of the twentieth is coming to an end.

Ehrenhalt, The Great Inversion, 2013

1 Introduction

Deep seated poverty marked US inner cities for most of the post-WWII era. Towards the end of the 20th century, however, the storyline changed. In 1980, 2-3 bedroom, single-family residences in major US cities were more expensive outside than inside the 10-mile radius ring, reflecting the then dominant urban pattern – affluent suburbs alongside inner cities in a seemingly unbreakable tailspin. Fast forward a generation and inner-city real estate has regained its footing. In fact, the price-distance pattern reversed (Figure 1). The ascendancy of the city center forms the set piece of what is loosely referred to as gentrification. Manifest already in 1990, it has continued unabated since – the 1980s crime wave and the recent housing correction notwithstanding.

Why this sea change? Why has gentrification replaced suburbanization as the urban trend of note? Why is it that demographic groups that formerly would have headed straight to the suburbs now seek to stay in the city, e.g., Couture and Handbury [2015], Baum-Snow and Hartley [2016], Hwang and Lin [forthcoming]?

One possibility is that the higher demand for centrally located housing stems from the growth in the number of high-income individuals, some of whom always favored central-city living [Gyourko et al., 2013]. Between 1970 and 2010, the US population grew by more than 100 million people and the urbanization rate increased from 73 to 81 percent.[1] Greater demand combined with land scarcity results in upward price pressure in the city center. But inelastic supply is not limited to city centers. Many upscale towns effectively limit housing supply by local ordinances and the like [Ortalo-Magne and Prat, 2014], but price increases have been markedly concentrated around city centers (cf. Figure 1).[2]

[1] See Davis and Heathcote [2007] for the rising share of land rent in housing values.

[2] For instance, while real estate prices in Manhattan rose by 33% in the last 10 years, they shrunk in

More congested roads is another "more people" related explanation. Congestion makes commuting more time consuming, less predictable, and less convenient.[3] Glaeser et al. [2008] argued that the concentration of poverty in the central city could be explained by the availability of public transportation, which is cheap but slow compared to commuting by car. Car ownership, higher among the rich, explains why the rich would locate away from the city center. If faced with only one mode of transportation, effectively the case if car travel is not faster than public transportation, then the rich would prefer to locate in the city. However, a number of large US cites actually experienced a population decrease in the last decades (e.g., Detroit) and the ascendancy of the city center evident in Figure 1 holds both for cities that grew and cities that shrank.[4]

The life cycle of real estate may be another factor driving urban trends. If high-income households prefer a younger housing stock, then that could explain why they headed to the then newly developed suburbs. The subsequent return to the city could similarly be explained by the arrival of new construction as the old real estate reached the end of its life span [Brueckner and Rosenthal, 2009]. This explanation, however, ignores the extensive renovation and residential conversion of urban real estate that has taken place in the wake of gentrification, e.g., Helms [2003], which suggests that in the US, high-income households have picked their preferred place of residence and then built or renovated existing real estate to their specifications.

Decline in crime is another candidate explanation: high crime rates have been found to contribute to urban flight [Cullen and Levitt, 1999], and it stands to reason that the drastic decline in crime since the early 1990s would have the reverse effect.[5] Ellen and O'Regan [2010] extended Cullen and Levitt's analysis to cover the 1990s, and thus studied periods of both rising and declining crime. They found only a limited role for crime, however, consistent with our observation that the price-distance relationship pivoted already in the 1980s, a decade in which violent crime rose [Donohue and Levitt, 2001]. We also note that European cities did not experience a level of crime in their inner cities anywhere near US levels but have nonetheless seen similar, core-centered, urban renewal, e.g., Carpenter and Lees [1995], Boterman et al. [2010]. Taken together, these observations suggest that while lower crime has made the central city more livable, the sources of gentrification may be elsewhere.

Yet another strand of the literature has focused on the role of local amenities. For instance,

suburbs such as White Plains and Chappaqua, both affluent towns in the NYC metro area (`http://www.neighborhoodscout.com/`).

[3]The daily commuting time has rise by about eight minutes since 1980 [McKenzie and Rapino, 2011].

[4]See Appendix Figures A1 and A2.

[5]The origins of crime's rise and fall are debated. A non-exhaustive list includes the removal of school prayers, abortion legalization [Donohue and Levitt, 2001, Foote and Goetz, 2008], more aggressive and targeted policing, greater incarceration rates, electronic surveillance, ATMs and credit cards, the crack epidemic, and environmental factors such as lead exposure [Reyes, 2007].

a concentration of cultural institutions, parks, and monuments boosts the attractiveness of the central city and European cities' advantage in that department may explain why Europe did not see US-style suburbanization [Brueckner et al., 1999]. Clearly, this explanation speaks more to post-WWII suburbanization than the urban revival of late. Furthermore, although the US does not have Paris, it does have New York City. By the end of the 19th century, New York City was the financial and cultural capital of the world's greatest economy, and it shows in the many cultural institutions in place well before the post-war decades of white flight and urban decay.[6]

Time varying amenities are better placed to account for gentrification. A caveat, however, is that such amenities are likely endogenous to the socio-economic characteristics of the local population, e.g., Albouy and Lue [2015]. The "spreading" through space – one location's good amenities boosting the attractiveness of nearby areas and helping them attract a more affluent demographic in a self-reinforcing cycle – can provide a handle on endogeneity [Guerrieri et al., 2013]. Still, the question of ground zero remains.

In this paper we focus on one local amenity – centrality – a feature which by and large has been fixed.[7] What changed, we argue, is its importance. The roots of this change, we propose, can be found in the greater labor supply of high-income households. For instance, in 1980, some 60 percent of prime working age households[8] at the 80th percentile of the income distribution had at least one householder (spouse or head) who did not work full-time. By 2010, that figure was down by a third (see Figure 2).

Individual labor supply by education paints a similar picture. The share of the college educated who work full time has increased and the increase has been particularly pronounced for college women and for those with advanced degrees. Further, among the skilled, the 50+ hour week (or "long hours," to use the terminology of Kuhn and Lozano [2006]) has become more common (see Figures 3 and 4).[9]

Greater labor supply, we propose, has moved high-income households towards the city center for primarily two reasons. First, longer hours have been accompanied by a reduction in leisure for the skilled [Aguiar and Hurst, 2009, table 2-2], a scarcity that would push towards different time use. By locating close to work, the commute can be shorter (in minutes) and/or more enjoyable. When work is in the center, nearby residence affords more options for the

[6] In 1975, New York City came very close to filing for bankruptcy.

[7] Clearly centrality itself is the result of a confluence of factors, but these were largely settled before the current wave of gentrification.

[8] Defined as households where at least one householder (head or spouse) was of prime working age (25-55).

[9] Use of the Current Population Surveys (CPS) allows for finer education categories (but limits the time period). Distinguishing between just college and a graduate degree, the skill-hours gradient still holds. For men, the difference is particularly pronounced at the 50+ hour margin, while for women it is evident also at the 40+ hour margin, Appendix Figures A3 and A4.

daily commute, notably rail transit and walking, modes of transportation that excel over short distances.

Second, land use models tell us that the downside of centrality is higher cost of space. However, long work hours may reduce the lure of the suburban space-advantage. With little time to host dinner parties or engage in hobbies, the home no longer need to cater to such extra curricular activities. Instead, the home's primary function reduces to that of basic restoration, in the limit, a place to sleep.

Long hours do not deny that local amenities affect gentrification, but we view them as amplifiers, not drivers.[10] Greater purchasing power in high-density areas paves the way for cultural and recreational amenities such as galleries, performance venues, restaurants and bars, etc., thus strengthening the attractiveness of the city center [Glaeser et al., 2001, Couture and Handbury, 2015].

Longer hours by skilled workers is an empirical fact, not an inevitability. High wages and low hours could go together. Why is it that the substitution effect has dominated the income effect? The answer may lie in the rise in the returns to skill [Katz and Murphy, 1992, Juhn et al., 1993, Autor et al., 2008] ushered in by the Information Technology (IT) revolution [Greenwood and Jovanovic, 1999, Hobijn and Jovanovic, 2001]. Beyond college or graduate school, most skill building takes place on the job in the form of working, a possibility consistent with long hours being highly rewarded, e.g., [Goldin, 2014]. Thus our paper is also related to the growing literature on the spatial and housing market consequences of the IT revolution, e.g., Berry and Glaeser [2005], Beaudry et al. [2010], Moretti [2013], Autor and Dorn [2013].

To investigate the empirical content of our hypothesis, one approach would be to regress housing prices in a locality on the presence of full-time skilled workers in that locality, as well as a battery of controls. However, the weaknesses are obvious. Full-time skilled workers have high incomes. That high-income households also consume more housing (in dollar terms) would be neither surprising nor speak to our hypothesis. The central city could have been more attractive to the high-skill-low-leisure population for reasons unrelated to long hours. For instance, as mentioned, the return of high-income households could be driven by the cultural amenities on offer in the city center, a pendulum swing springing from the real-estate life cycle, or simply a cultural reaction against the ubiquity of subdivisions and cul-de-sacs (e.g., Gallagher [2014]). Reverse causality is also a possibility – higher rents prompting greater labor supply [Johnson, 2012].

Our empirical strategy is to use a Bartik-type demand shifter [Bartik, 1991]. This demand shifter uses a location's employment composition in a base year and national employment trends

[10] Diamond [2016] found that greater city level demand for skilled labor reduced crime.

excluding the locality in question to generate predicted labor demand. Considering the likely role of the IT-revolution in the rising returns to skill, we chose 1970 as our base year. The exclusion of the focal locality allows for the generation of arguably exogenous variation in the demand for full-time skilled labor, which in turn result in housing demand close to those jobs, viz. the city center.

Our principal data set draws from restricted-use decennial census micro data or the American Community Survey (ACS) and covers the period 1980-2010. The 1980 census is the first year for which we have a Bartik shifter (1970 being the base year). The last year is dictated by data availability (2010 short hands for the pooled 5-year ACS sample 2008-2012). We focus on top-20 cities by population size, a list that has not been stable over the study period. We include cities that were in the top-20 in either 1970 or 2010, the case for 27 cities. We aggregate micro-level data to the census tract, and our key dependent variable is the median price for a two- or three-bedroom home in the tract. To measure the presence of full-time skilled workers, we consider two cut offs for full time and skill: 40 or 50 hours per week and a four-year college or advanced degree, respectively.

Our endogenous variable of interest varies at the sub-metro level. Therefore, we interact the city-year demand shifter with the (tract) distance from the central business district (CBD) for a shifter that varies within each city-year. In recognition of the role of the IT revolution in driving spatial variation in demand for skilled labor [Beaudry et al., 2010] we use 1970 as the base year and construct the demand shifter from a combination of 1970 city-level employment and national growth trends, following a number of recent papers, e.g., Diamond [2016], Moretti [2013].

Our baseline specification includes city-year and city-distance fixed effects. City-year fixed effects absorbs any city-wide changes, for instance city-level labor demand shifts or credit expansion,[11] as well as any city fixed effects, such as topography, or climate, or any historically fixed factors such a history of racial tension or pre-existing infrastructure, monuments, or institutions. City-distance fixed effects are similar to tract fixed effects (as a robustness check, we cross walk tracts, which allows us to include tract fixed effects). Our most demanding specification also include distance-year fixed effects, which absorbs any overall changes to the distance-price relationship.

For starters, we first establish that housing prices correlated positively with the Bartik demand shifter and the relationship is stronger close to the CBD. Further, the Bartik shifter correlates positively with the fraction of the population that is skilled and attached to the labor force, as measured by education level and hours worked.

[11] For recent papers on the role of the latter, see Glaeser et al. [2010], Favara and Imbs [2015].

In order to investigate the role of confounders, we look at sub-samples of the data. To check for the role of lower crime we cut the data in two ways: (i) by year – the decline started in the early 1990s; and (ii) by the extent to which crime declined (cities with high declines and cities with low declines or even increases). Another cut of interest is cities that grew and cities that shrank (since 1970). The role of congestion, crowding of high income individuals, or foreign demand is presumably limited in shrinking cities (viz., Detroit, St. Louis, Cleveland, Baltimore). New York City is unique in many ways (the largest US city, the financial sector, foreign buyers, etc). However, our results are robust to its exclusion. Throughout, we find a strong positive effect of the Bartik demand shifter close to the CBD.

Other tract demographics such as race, income, or marital status, moved with the Bartik shifter, and in the expected direction. However, we view these changes as incidental, derived from the primary relationship between the Bartik shifter and the educational and employment profile of the tract population.

Turning to the relationship between full-time skilled workers and housing prices, we find empirical support both in OLS and IV regressions (the latter employing the Bartik demand shifter). Skilled full-time workers not only locate close to the CBD, their presence raises housing prices. The implied price changes suggest that the growth of skilled full-time workers captures the 1980-2010 housing-price changes well.

Lastly, our argument for the rising salience of centrality rests on the assumption that skilled jobs are predominantly located in the city center and we confirm this to be the case (unskilled jobs, on the other hand, show less geographic concentration, and increasingly so).

The remainder of the paper is organized as follows. We round out this section by sketching a conceptual framework to help fix ideas. Section 2 describes our empirical strategy and the data. Section 3 presents our results. Section 4 concludes.

1.1 Conceptual framework

This sections sketches a theoretical framework in which (exogenously given) hours determine location choice and thus by extension, housing prices. When the rich need to work long hours, they prefer to live in the city, whereas at shorter hours, the suburb may be preferred. When the rich prefer the suburb, prices rice with distance to the city center, and vice versa when the rich prefer the city. The main innovation on the canonical land use model is thus to consider the role of work hours on location choice, a minor extension that can be skipped without loss of context.

Our premise is the canonical mono-centric model of urban development in which commuting time to the suburb is weighted against the lower cost of space [Mieszkowski and Mills, 1993,

Rappaport, 2014, 2016]. There are two locations, city and suburb, indicated by subscripts $i = c$ and $i = s$, respectively. All jobs are located in the city whereas both locations can house residential housing stock. Since all jobs are in the city, the suburb comes with a commute. For simplicity we assume that

$$t = \begin{cases} t_s > 0 & \text{if } i = s, \\ 0 & \text{otherwise.} \end{cases}$$

Consider a population of workers, half of whom are skilled and command a high wage w, half of whom are unskilled and command a low wage (we do not need a notation for their wage). For brevity, we will refer to them as rich and poor, respectively. Workers have 16 hours per day at their disposal. Time can be used in three ways: work, h, commuting, t, and leisure, l. To focus on the role of longer hours on location choice, everybody works and we do not allow workers to choose their hours. In line with stylized facts for the period concerned, we assume that the hours for high-wage workers are the same or longer than for low-wage workers.[12]

Construction costs are such that only the rich can afford to build durable housing. The poor either live in already built housing or in cheap non-durable housing and for the purpose of this section, housing will refer to the durable type. Housing accumulate as location specific housing stock, $s_i \geq 0, i = c, s$ which rents at $r_i, i = c, s$ (per square foot) whose floor is given by the cost of maintenance. For simplicity, we assume maintenance to be the same in the city and the suburb, $m_c = m_s = m > 0$.[13]

Initially, only the city is developed. If aggregate demand by the rich, s_i^*, at rental price $r_i = m$, exceeds the available housing stock, then housing gets built to meet demand. Land is assumed in infinite supply in the suburb and thus the marginal cost of new construction consists of labor and material, both which are elastically supplied and thus the marginal cost of construction is a constant $c_s > m$. In the city, land is scarce and we model this by letting the marginal cost of construction depend positively on the housing stock $c_c(s_c), c_c(0) = c_s > m, c_c'() > 0$.[14]

Rich workers rent a_i square feet at the rental price r_i, in either the city or suburb. The rent is

$$r_i = \begin{cases} m & \text{if } s_i^* \leq s_i, \\ c_i & \text{otherwise, } i = c, s; \end{cases}$$

[12] If low wage workers worked longer hours, on the other hand, then it is possible that low wage workers could outbid high wage workers to shorten their commute, but that is not the empirically relevant case.

[13] We allow for housing supply in order to demonstrate robustness to this dose of reality. Qualitatively similar, but amplified, results are obtained if the housing stock is fixed.

[14] For instance, from higher land rent and construction costs.

recall that the suburban construction cost, c_s, is a constant whereas the city construction cost, c_c, varies positively with the existing housing stock s_c.[15]

Workers derive utility from living space, a, leisure, l, and a numeraire consumption good, x. We let utility be of Cobb-Douglas form:

$$u = x^{\alpha} a^{\beta} l^{\gamma},\ \alpha, \beta, \gamma > 0.$$

The budget constraints are:

$$x + r_i a_i \leq w\, h,$$

and

$$l = 16 - h - t.$$

The optimization problem may thus be viewed as a two-stage process: first determine which locality would give the highest utility, home size chosen optimally at the implied rental price; then pick the location that gives the highest utility. In other words, to pin down the urban location pattern, we need to know where the rich obtain the highest utility: in the city or the suburb? Let a_i^* denote the utility maximizing home size in locality i at the implied rental price r_i. The city gives higher utility if

$$\left(\frac{wh - r_c a_c^*}{wh - r_s a_s^*}\right)^{\alpha} \left(\frac{a_c^*}{a_s^*}\right)^{\beta} > \left(\frac{16 - h - t_s}{16 - h}\right)^{\gamma}. \tag{1}$$

Note that if the housing stocks and construction costs results in suburban homes being relatively large (low a_c^*/a_s^*), for instance from abundance of undeveloped land and cheap labor, the rich favor the suburbs (possible contributing reasons for why the US suburbanization patterns did not happen in Western Europe).

Now, how do longer hours affect location decisions? The right hand side of 1 goes to 0 as $h \to 16 - t_s$. The left hand side has two components: the city-suburb consumption ratio, which under our functional form assumption equals 1,

$$\frac{wh - r_c a_c^*}{wh - r_s a_s^*} = \frac{x_c^*}{x_s^*} = 1; \tag{2}$$

and the city-suburb optimal housing size ratio

$$\frac{a_c^*}{a_s^*}. \tag{3}$$

[15]Strictly speaking, there is an intermediate case where demand at the cost of maintenance exceeds the existing housing stock but demand is not strong enough to elicit new construction. In that case, rent is indeterminate in the range (m, c_i). This possibility does not change our qualitative results.

The ratio of city to suburban home sizes decrease with higher hours because hours raise income and with higher income, more money is spent on housing and the suburbs have more elastic housing supply.[16] However, $\frac{a_c^*}{a_s^*}$ remains above zero, thus higher hours, at least in the limit favor the city.

For the same reason, a higher wage reduces the left hand side of inequality 1, and thus makes it more likely that for any given number of hours worked, the suburban location is favored. Thus, high(er) wages may have contributed to the US suburban expansion in the post WWII period. Note, however, that the city will be favored once the right hand side approaches zero, the case for long commuting times and/or high working hours.

Consider the following three stylized phases of US urban development: (i) the pre-automobile city; (ii) automobile-enabled suburban expansion; and (iii) urban renewal. The role of the automobile in making the transition from Phase (i) to Phase (ii) is widely recognized. In Phase (i) rich and poor live in the city, the rich in durable housing and the poor in non-durable housing, the suburb is undeveloped. In Phase (ii) the rich have the option to move to the suburb, which they do if work hours h are low and the houses they can afford to build are sufficiently large to compensate for the commute (a_c^*/a_s^* is low enough). The poor stay in the city in the housing vacated by the rich and pay rent $r_c = m$. Clearly, the square foot rent is higher in the suburb than the city (maintenance is cheaper than new construction, $m < c_c(0) = c_s$).

In Phase (iii), longer working hours h (or longer commute t_s) make the rich return to the city. If aggregate demand (by the rich) is such that new construction is induced, then rent in the city, $r_c = c_c(s_c^*)$ is greater than rent in the suburb $r_s = m$ (paid by the poor who move there).[17]

Individuals v. Families We have focused on individual labor supply, departing from a long tradition of treating labor supply and location decisions in the context of a family [Oi, 1976, White, 1977, Mincer, 1978, Madden, 1980, Madden and White, 1980, Costa and Kahn, 2000, Compton and Pollak, 2007].

The distinction between the individual and the household becomes moot with positive assortative matching and low degree of intra-household specialization, two trends that arguably have gained currency.[18] While it is still true that women's labor supply is more sensitive to

[16] The optimal housing size, a_i is given by the implied housing stock from the purchasing power of rich households, divided by their numbers.

[17] So far, the displacement of poor households in gentrifying neighborhoods appear to have been modest, suggesting that gentrifiers occupied new construction or rehabilitated unoccupied units [McKinnish et al., 2010, Ellen and O'Regan, 2011], as might be expected from strong protection of renters' rights. Moreover, the lack of displacement, may have contributed to the polarization of incomes in the city as poor incumbents and newcomers rich enough to afford new construction edge out middle income households.

[18] For empirical evidence on sorting, see Juhn and Murphy [1997], Heim [2007], DiCecio et al. [2008], McGrattan and Rogerson [2008], Schwartz [2010].

commuting cost [Black et al., 2014] and women tend to work more locally, increasingly, the commuting patterns of women are approaching those of men [Edlund et al., 2015, figure 9].

To see how our framework would handle couples and singles, let us assume positive assortative matching and equally many men and women in each skill group. Furthermore, individuals can be married or single. Singles work, but a fraction of married women do not.

High-wage households are of particular interest and we have three groups to consider: singles, dual-earner couple, and breadwinner-housewife couple. On a per capita basis, the first two household types are similar – they have high earnings and little leisure (compared to the breadwinner-housewife household). In this case, heterogeneity among skilled household could take the following form: singles and dual-earner households locate in the city; the breadwinner-housewife couple locate in the suburb. If breadwinner-housewife couples dominate among the skilled, the singles and dual-earner couples would not fill up the city and rents would remain low, allowing others to reside there as well. As the number of singles or dual-earner couples among the skilled increases, low-skilled households are priced out.

The addition of children – non-earning members with space requirements – would push households towards the suburb reinforcing the above tendency of the breadwinner-housewife household to favor the suburb.

To sum up, the model points to the role of work hours in shaping optimal location choice. At short hours, the affordability of housing in the suburbs dominate (once commutable thanks to the automobile), whereas at long hours, the additional space cannot compensate for the (scarce) leisure lost to commuting.

2 Data

Our primary data set is drawn from the decennial censuses of 1980, 1990 and 2000, and the American Community Survey (ACS) pooled 5-year sample for 2010 (years 2008-2012). The chosen level of aggregation is the census tract. The Census wrote: "Census tracts generally have a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people."[19]

Our housing-price measure is based on self-reported values of owner-occupied two-to-three bedroom single family homes.[20] We use households no more than ten years in the current residence on the assumption that owners of more recently transacted units would be more knowledgeable about the going market price.[21]

[19] `http://www.census.gov/geo/reference/gtc/gtc_ct.html`

[20] Two and three bedroom homes are modal, for new construction see `https://www.census.gov/construction/chars/pdf/soldbedrooms.pdf`

[21] Including all households yields similar results.

To capture the rise of the high-income-low-leisure demographic, we focus on full-time workers with a college degree, ages 25 to 55. This age group captures prime working ages, the years after college completion but before retirement concerns; it is also a key home-buying age group.[22]

Further details on data set and variable construction are in the Data Appendix.

2.1 Sample cities

Geographically, we limit our sample to cities that were in the top-20 (population wise) either in 1970 or 2010. There were 27 such cities which for simplicity we will refer to as "top-20." The list was topped by New York City in both years. Phoenix and Memphis claimed the number 20 spot in 1970 and 2010 respectively.

We used the tract centroid location to calculate the distance to the CBD (Appendix Table A8 lists the cities and the respective CBDs). We include tracts within 35 miles of the CBD whether administratively part of the city, or not. This geographic restriction is arbitrary but arguably delineates a commutable area, the outer reaches of which would be about a one-hour commute to the CBD. We have about 65 thousand tract-year observations, or an average of 600 tracts per city and year.[23]

2.2 Bartik-type skilled labor demand shifter

We used the public use version of the decennial censuses (IPUMS) to calculate the city-specific demand shifter for skilled employment from the national growth rates of employment of college workers in industry h, excluding the city in question, $-j$, weighted by each industry's 1970 employment share in city j. The base year 1970 was chosen because the rise in returns to skill dates to the IT revolution that started in the ensuing decade [Beaudry et al., 2010].

Specifically, focusing on ages 25-55, we construct our Bartik demand shifter for skilled labor demand for city j and year t as:

$$Z_{jt} = \frac{1}{N_{j,1970}} \sum_{h}^{41} n_{h,j,1970} \times (\ln n_{h,-j,t} - \ln n_{h,-j,1970}), \tag{4}$$

where

$N_{j,t}$ is the number of workers in city j and year $t = 1980, 1990, 2000, 2010$,

$n_{h,j,t}$ is the number of college educated workers in industry h, city j, year t,

$n_{h,-j,t}$ is the number of college workers in industry h and year t, excluding city j. In other

[22] National Association of Realtors [2014].

[23] For disclosure reasons, the sample size is rounded.

words, the first factor is the 1970 city specific industry share in 1970; the second factor is the logged national growth, excluding city j, for that industry in terms of college workers. Thus, Z_{jt} can be interpreted as the share of employment in city j predicted to be held by college educated workers.

Throughout the study period, Z_{jt} is book-ended by San Antonio and Washington DC. In 1980, San Antonio's predicted growth of college educated workers was 9 percent; the number for DC was 23 percent. In 2010, those numbers were 18 and 46 percent respectively.

The Bartik measure is city specific (see Table A8).[24] To allow the demand shifter to propagate differentially throughout the city we interact it with a measure of the tract distance from the CBD. We expect the demand shifter to operate more strongly close to the CBD where skilled jobs are concentrated (confirmed by data to be the case).

2.3 Descriptives

We start by presenting descriptive statistics (means) by year and distance from the CBD. We present these descriptives in tabular form where the data is grouped in four distance bins: 0-3, 3-10, 10-20, and 20-35 miles away (Table 1). For finer parsing by distance, we present smoothed polynomials based on 1-mile distance intervals in a series of graphs.

House prices Between 1980 and 2010, the median price for a two-or-three bedroom one-family residences in the top-20 cities rose by 30 percent from 92.5 to 120.5 thousand 1980 dollars (or from 267 to 348 thousand 2016 dollars). Turning to price changes by distance from the CBD, we see that price increases were higher in more centrally located tracts. In the core (0-3 miles), prices more than doubled. In tracts 3-10 miles out, prices rose by 60 percent, whereas price increases were a mere 10 and 6 percent in tracts 10-20 and 20-35 miles out, respectively. In fact, the price profile flips. In 1980, prices in the periphery were 50 percent higher than in the center. By 2010, it is prices in the center that are higher, by about 40 percent (Table 1 and Figure 1).

Full-time skilled workers We employ the following notation: BA-, less than four-year college; BA, four-year college but no advanced degree; MA+, advanced degree; and BA+ refers to BA and MA+ combined. Let $FT(h, e)$ denote the fraction of adults 25-55 with education $e, e = BA+, MA+$ and work weeks exceeding $h, h = 40, 50$ hours.

The fraction of adults 25-55 who worked full-time and had a college degree increased in the sample cities throughout the study period, a development that was particularly pronounced for

[24] Except Dallas-Fort Worth, coded as one city in the IPUMS.

women (Table 1 and Figures 5 and 6).[25]

The period also saw a distinctive improvement in the center's ability to attract this better educated and more employed population. In 1980, $FT(40, BA+)$ was higher outside than inside the 10-mile perimeter. By 2010, $FT(40, BA+)$ had risen markedly in the city core (0-3 miles), making up a third of the core's population (compared to a quarter overall). The shift towards the city center was even more pronounced among those working long (50+) hours (Table 1).[26]

Breaking down residence location by gender, we see that full-time skilled women consistently favored the city core, and more so the longer the hours worked and the higher the degree (Figure 6). The main difference over time is the growth in their numbers. Full-time skilled men, on the other hand, show more willingness to live away from the CBD and this is particularly pronounced at the lower skill and hour cutoffs (Figure 5). Over time, however, men, in particular $FT(50, MA+)$ men, approach women in choosing central city location (possibly because many of them are matched to women with not just similar education but also similar hours).

Location of jobs, distance from the CBD Figure 7 shows the distance and skill distribution of jobs by year. The figure shows the fraction of jobs held by a given education group (BA-, BA, BA+, MA+) at a given distance from the CBD (among all jobs held by adults 25-55, unconditional on hours, within 35 miles from the CBD, in a given year). We see that in 1980, skilled and unskilled jobs were concentrated in the city core. By 2010, however, there has been pronounced decline of unskilled jobs inside the 10-mile ring. Skilled jobs, by contrast, maintain their concentration close to the CBD, a concentration that is particularly pronounced for jobs held by workers with a graduate degree.

Age composition We focus on full-time skilled workers, ages 25-55. It is possible that this framing leaves out important demographic developments. Has gentrification not been associated with hipsters and empty nesters?

Table 1 reports the population shares by age group and distance category and we see that the shares of the old and the young (including children) have declined overall, whereas the prime working age group increased by 13 percent (from 40 to 45 percent). Furthermore, the increase was concentrated in the 0-3 mile core, where the share went from 39 to 50 percent over the study period. These patterns are further illustrated in Figure 8. In 1980, the 25-55 age group's presence was j-shaped, bottoming out around 4 miles from the CBD. In 2010, the j had fallen on its back – the prime working ages dominate the city center and decreased monotonically thence.

[25] This pattern also held nationally (Figures 3, 4, A3, and A4).
[26] Meanwhile, unskilled full-time workers moved away from the center (Figure A5).

While young adults, 19 to 24 years old, have long favored the central city, their presence decreased over the study period. Thus, this demographic does not seem to be a prime driver of gentrification, perhaps not surprising considering their limited purchasing power relative to the 25-55 age group.

As for ages 56 and up, this group has grown overall but the growth is concentrated 10 miles out of the CBD. Within 10 miles of the CBD, there is a decrease. This was true of both the 56-65 and the 66+ age groups, although it was more pronounced for the latter (Table 1).

Other demographics Over the period, income (total personal, 1980$) rose by 25 percent, and perhaps unsurprisingly, the gains were concentrated in tracts close to the CBD. In the 0-3 mile core, income rose by almost 80 percent. The percent non-Hispanic white declined by 12 percentage points, whereas the percent blacks increased only slightly (from 19.4 to 19.6 percent). In the central city, however, the pattern was reversed: percent black declined while percent non-Hispanic white held steady. As for marital status, marriage declined overall and in percentage terms the decline was steeper within 10 miles of the city center (25 v. 15 percent), Table 1.

3 Analysis

We start by investigating the reduced form effect of the skilled labor demand shock on housing prices by estimating a regression of the following form:

$$PRICE_{ijdt} = \beta_0 \times Z_{jt} + \boldsymbol{F'_{ijdt}}\boldsymbol{\beta_1} \times Z_{jt} + \alpha_j + \alpha_d + \alpha_t + \epsilon_{ijdt}, \tag{5}$$

where $PRICE_{ijdt}$ is the housing price in tract i at distance d from the CBD of city j in year t, Z_{jt} is the exogenous labor demand shock in city j in year t (see equation (4)) and $\boldsymbol{F_{ijdt}}$ is a $m \times 1$ vector with functions of $dist_{ijdt}$ – tract distance from the CBD – and $\boldsymbol{\beta_1}$ is the conformable vector of parameters. We allow for distance differential impacts of Z_{jt} through $\boldsymbol{F'_{ijdt}}\boldsymbol{\beta_1} \times Z_{jt}$. Our coefficient vector of interest is $\boldsymbol{\beta_1}$, and our hypothesis is that Z_{jt} has a greater impact on prices close to the CBD.

To capture this distance differential effect, we consider three specifications of $\boldsymbol{F_{ijdt}}$:

$$\boldsymbol{F_{ijdt}} = \begin{cases} D3_d = (d1_d, d2_d, d3_d)' & (m = 3) \quad \text{or;} \\ d1_d, & (m = 1) \quad \text{or;} \\ (dist_{ijdt}, dist^2_{ijdt})' & (m = 2), \end{cases}$$

where $dk_d, k = 1, 2, 3$ indicates the 0-3 mile core, the 3-10 mile radius ring, and the 10-20 mile

ring respectively (the 20-35 mile ring is the reference category). These rings were chosen based on preliminary inspection of the data.sw The third specification allows the distance differential effect to be a quadratic function of distance.

City, distance and year fixed effects are captured by α_j, α_d and α_t. Distance fixed effects control for factors such as topography or geology. We discretize the distance fixed effects by using the three distance dummies in $D3_d$. We also consider two-by-two interactions: city-year, city-distance and distance-year fixed effects $(\alpha_{jt}, \alpha_{jd}, \alpha_{dt})$.

City-year fixed effects parcel out time effects common to tracts in a city, for instance, city-wide changes to policing policies.

Tract characteristics can change over time (e.g., endogenous amenities). Distance-year fixed effect account for time evolving characteristics that are related to the location of the tract relative to the CBD. The distance-year fixed effects correspond to year-specific effects in $D3_d$. The choice of $D3_d$ for the distance-year fixed effect is motivated on the grounds of parsimony, the intervals chosen to capture shared price dynamics.[27]

We would like to control for tract characteristics, for instance by including tract fixed effects in equation (5). However, tracts change between the censuses. While possible, the construction of a tract panel data set introduces measurement error and therefore we favor the repeated cross-sectional data for the main analysis.[28] The city-distance fixed effects consider city-specific effects in $D20_d$, a vector of 20 distance dummies, $rk_d, k = 1, 2, ..., 20$ indicating that the tract is [0,1), [1,2),...,[19-20) mile $r$ing. These city specific 1-mile ring fixed effects may be considered substitutes for tract fixed effects.

The inclusion of the full set of fixed effects means that much of the city-year-distance level variation is absorbed – leaving little, but more exogenous, residual variation for recovering our main parameter vector of interest $\boldsymbol{\beta_1}$.

Throughout, we cluster standard errors at the city level and weigh tracts by their population ages 25-55.[29]

3.1 Bartik-type skilled labor demand shifter on housing prices

Table 2, column 1, shows the results from simply regressing $PRICE_{ijdt}$ on the demand shifter Z_{jt}, city, distance, and year fixed effects. We see that higher Z_{jt} is associated with higher housing prices: 1 unit of Z_{jt} is associated with an additional \$522k. The average change in Z

[27] Results were similar for finer distance-year fixed effects, therefore the more parsimonious specification was chosen.

[28] We will show that results hold including tract fixed effects using a constructed panel data set in the robustness analysis.

[29] Regression results not using weights were similar.

was 0.128 and thus the implied price change is about \$70k (out of an observed difference of \$88k, Table 1, 1980 constant dollars).

The average effect of the demand shifter Z_{jt} masks substantial heterogeneity by distance as can be seen in column 2. The effect in the 0-3 mile core is more than twice as large as the effect in the 3-10 mile ring (588.9/234.5=2.51), and more than six times the effect in the 10-20 mile ring. In this specification, the effect beyond 20 miles is given by the main effect on Z_{jt}. While positive, it is not statistically significant at conventional levels. In sum, we find a positive effect of the demand shock and the effect is concentrated close to the CBD.

Columns 3-5 sequentially introduce city-year, city-distance and distance-year fixed effects. These fixed effects partial out confounders that could be correlated with the demand shock and vary at the corresponding level. Column 3 replaces the city- and year fixed effects with the more flexible city-year fixed effects (the main effect of Z_{jt} can no longer be recovered since it varies at the city-year level) and we see that the results are similar. Column 4 replaces distance fixed effects by city-distance fixed effects and, again, results remain similar.

Column 5 adds distance-year fixed effects, an inclusion that renders the coefficient on the demand shock on the 3-10 mile ring ($d2$) statistically insignificant, but not statistically different from the effect on the 10-20 mile ring ($d3$). The effect on the city core ($d1$), however, is slightly stronger. Therefore, in column 6 we keep the full set of fixed effects but let the effect of the demand shock vary by whether in the 0-3 mile core or not – we find that the city-core effect remains.

In Columns 7-9 we let the distance differential effect of Z_{jt} be captured by Z_{jt} interacted with continuous distance squared and introduce city-year, city-distance and distance-year fixed effects sequentially. As expected, the effect of the demand shock decreases with distance but at a decreasing rate. The introduction of more demanding fixed effects reduces the coefficient estimates, but they remain statistically significant throughout.

As a robustness, we also estimate the above series of specifications using logged housing prices (Appendix Table A1).[30] We see similar results, the main exception being that results including distance-year fixed effects only survive in the specification where the distance-differential effect enters squared (column 6).

Overall, results remain qualitatively the same under the alternative fixed effects considered – reassuring evidence that the demand shock is mostly unrelated to other city-year, city-distance and distance-year varying characteristics. Results are also robust to alternative distance-differential effects in $\boldsymbol{F_{ijdt}}$. Henceforth, we focus on the specification using $(d1_d, d2_d, d3_d)$ or $d1_d$ in $\boldsymbol{F_{ijdt}}$ as they capture distance-differential effects using a parsimonious spline.

[30] Since most homes are purchased as primary residences, the primary driver of willingness to pay is the user value rather than the return on investment. Therefore, we favor prices in levels.

Since our posited mechanism for rising home prices hinges on high-income households working longer hours, we now turn to investigating the effects of the skilled-labor demand shock on tract demographics.

3.2 Bartik-type skilled labor demand shifter on demographics

We start by investigating how our key demographic – full-time skilled employment – responds to the skilled-labor demand shifter. We let full time be 40 hours or more per week and consider two measures of skill: (1) college or more, $FT(40, BA+)$; and (2) graduate or more, $FT(40, MA+)$. (We also considered a higher cut off for full time, 50 hours or more, but since results are similar they are in the Appendix.)

Table 3, panel A shows the results for $FT(40, BA+)$, full-time workers with a college degree or more, and we follow the same specifications as in Table 2 but skip column 2 for brevity (that is, save for column 1, all specifications include city-year fixed effects).

We see that the demand shock has a positive (but insignificant) effect on our full-time skilled population measure (column 1), but there is important heterogeneity by distance: most of the effect is concentrated in the 0-3 mile core (column 2) and this result is strengthened as fixed effects are added to estimation (column 3 and 4). However, we also see that inclusion of the flexible fixed effects does not leave enough variation to identify effects outside the 0-3 mile core and we therefore drop the ring interaction terms to focus on the city core (column 5). The coefficient implies a change in fraction of skilled full-time residents of 16.5 percentage points (129.4×0.128), or 3/4 of the observed change over the period for tracts in the 0-3 mile core.

The results for full-time workers with a graduate degree or more, $FT(40, MA+)$, are in panel B. The implied change in fraction of skilled full-time residents in the 0-3 mile core is 7.4 (57.52×0.128) percentage points, or 80% of the observed change (column 5).

Two components are crucial for the relationship between tract housing prices and the skill and working hours characteristics of tract residents. First, skills are important because skilled workers have high wages and can bid up prices in their preferred locations. Second, skilled workers who face a long work week would want to avoid commuting. These two components, combined with skilled jobs being concentrated in the CBD, leads to our hypothesis that gentrification has resulted from a growing number of high-income households supplying more market hours. In order to investigate the importance of these two pieces, we consider two alternative measures of the local labor force.

First, we look at full-time workers with less than a college degree in column 1 of Table 4. Consistent with our hypothesis that this group would not be able to compete for (and possibly be less interested in) housing in the central city, we see that the estimated effect of the skilled

demand shock is negative (and borderline significant) in the city core whereas a positive effect can be found in the third ring ($d3$ or 10-20 miles out).

Second, we look at college men and women, regardless of hours worked. As expected, the demand shock has a weaker effect on their location choice. Working fewer hours, their willingness to live away from the CBD may be higher, column 2, Table 4.

We now turn to other tract characteristics that plausibly change with the Bartik demand shifter: percent black, white (non-Hispanic), married, and average personal income. These variables are correlated with the share of the population that is skilled and works full time, and we expect to find a relationship.

Columns 3-6, Table 4 show the results. As argued, some demographics move in tandem with the work characteristics of the residents. In particular, gentrification is linked to an inflow of high(er) income residents, and income and race are strongly correlated. Interestingly, we also see that gentrification is closely tied to (relatively) more married couples.

As argued before, a relationship is expected since education and employment status are correlated with these other demographics. The reason we favor $FT(\cdot)$ is the direct causal link from growth in a skill-intensive industry to a more skilled and employed population. That is, we view other demographics such as race, income, or marital status, as largely incidental or derived from this primary relationship.

3.3 Housing prices and full-time skilled work

We now turn to our hypothesis that the emergence of centrality as a top amenity is linked to greater labor supply of high-income households. We start by considering a regression of the following form:

$$PRICE_{ijdt} = \alpha_1 FT(h,e)_{ijdt} + \alpha_2 dist_{ijdt}\, FT(h,e)_{ijdt} + \alpha_3 dist_{ijdt} + \alpha_{jt} + \alpha_{jd} + \epsilon_{ijdt}, \quad (6)$$

where we will consider specifications where FT enters alone and where it is also interacted with distance. The interaction term captures possible differential effects of FT on housing prices by tract distance from the CBD. α_{jt} and α_{jd} are the previously defined city-year and city-distance fixed effects. We also consider adding distance-year fixed effects (α_{dt}) (on top of α_{jt} and α_{jd}).

In column 1, Table 5 we see the results for equation (6) when FT enters alone. Panel A shows the results for $FT(40, BA+)$ and the estimated coefficient implies an average increase in housing prices by \$21k (1.829×12.57). Column 2 adds distance-year fixed effects and the result remains virtually unchanged. Columns 3 and 4 allow for a distance-differential effect of

FT by adding an interaction term with distance, and we see that while of the expected sign and statistically significant (column 3), this result wanes once distance-year fixed effects are included (column 4), bringing the estimated effects on full-time skilled workers close to that without the interaction term (column 2). Panel B shows the results for $FT(40, MA+)$ and we see similar results (as were the results for $FT(50, BA+)$ and $FT(50, MA+)$, see Appendix Table A3).

3.3.1 Mechanisms

While the effects of FT on housing prices are surprisingly stable across specifications that control for a flexible set of fixed effects, it is still possible that our findings are driven by omitted variables correlated with both housing prices and tract demographics.

Our strategy for guarding against such confounding is to use a Bartik-type demand shifter for skilled labor. Under the assumptions that the Bartik demand shifter affects housing prices only through the posited channel and is orthogonal to any unobservable tract-by-year characteristics in the housing price equation, we can instrument our full-time skilled measure by the Bartik demand shifter in equation (6).

As the 'Starbucks effect' term suggests, gentrification has been accompanied by a change to local amenities. However, does the appearance of local services such as restaurants or retail establishments drive gentrification or do they merely reflect (anticipated) changes to the local demographics? If the latter, such tract-by-year changes need not pose a challenge to our identification strategy, they merely mediate effects of a more skilled and employed local population. "Permissible" indirect effects include changes in the retail landscape, law and order, transportation, quality of public schools, etc. that stem from the social, economic or political clout of a skilled, steadily employed population.

More worrying would be if demand growth for skilled labor affected other dimensions that in turn moved housing prices. For instance, a higher Bartik may result in higher tax revenues, which in turn allow for better policing, infrastructure, or civic initiatives that improve quality of life, and possibly more so close to the city center. Alternatively, a higher Bartik could also signal greater demand for office space, where higher prices reflect dwindling supply as residential real estate lose out to commercial such. Demand induced new construction (residential and commercial) would work as a counterweight, and the anecdotal evidence contained in the term "residential conversion" is born out by building permit evidence [Thomas, 2009, 2010, Ramsey, 2012]. Population statistics paint a similar picture, at least of late. Between 2000 and 2010, the central city population grew, reversing decades of population loss [Baum-Snow and Hartley, 2016].

Before moving to the IV results where we instrument FT using the Bartik demand shifter, we now turn to discussing alternative mechanisms.

Time-invariant tract characteristics While many amenities change over time, some are largely fixed. Examples include amenities related to topography or location (including centrality). To account for time-invariant tract effects, we construct a panel of tracts using cross walk files from US2010[31] in order to add tract fixed effects to equation 5. The cross-walk procedure is however such that there is a loss of fidelity to the underlying demographics. With this caveat in mind, Table 6, column 1, shows the results from adding tract fixed effects to the column 4 and 5 specifications of Table 2 in Panel A and B respectively.[32] Results are quite similar to those found in the cross sectional version of the data set, suggesting that the set of fixed effects, notably city-distance fixed effects, were passable substitutes for tract fixed effects.

Lower crime? Violent crime rose through the 1970s and 1980s but started to decline in the early 1990s. The decline in crime has been argued as a possible cause for gentrification. Therefore, we separate the years 1980 and 1990 from the years 2000 and 2010. This allows us to look at a period in which crime rose and one in which it declined.

Despite rising crime, the effects for the early years are similar to that of the whole period, Table 6, column 2. Results remain statistically significant through the inclusion of city-year and city-distance fixed effects (Panel A), but lose significance once distance-year fixed effects are added (Panel B).[33] Results for 2000 and 2010 in column 3 are stronger.

An alternative tack is to split the sample by the decline in crime. The city grouping was done with a view to split the sample in terms of number tracts. We group cities that saw large declines in crime and those that saw modest declines or even increases over the 1985-2012 period.[34] Qualitatively, results hold in both groups (Table 6, columns 4 and 5), but we also see that cities with large declines in crime saw a stronger response to the Bartik shifter.

Some of the cities with the steepest housing price increases have also seen the largest declines in crime, New York City being a case in point. New York City is also the largest US city and its prominence in finance, business, and culture means that its property market may also be singular. For instance, foreign buyers constitute a significant share of demand at the high end of the real estate market. For these reasons, we exclude New York City and find similar effects

[31] `http://www.s4.brown.edu/us2010/Researcher/Bridging.htm`

[32] City-distance fixed effects are absorbed by the tract fixed effects.

[33] Since the analysis breaks the sample by only two consecutive years (which are likely similar) adding the very demanding specification with both metro-year and distance-year fixed effects absorbs most of the variation of the instrument.

[34] The years for which crime statistics are readily available. `http://www.ucrdatatool.gov/Search/Crime/Local/TrendsInOneVarLarge.cfm`

among the remaining cities, Table 6, column 6.

These findings suggest that while falling crime is not behind the trend towards gentrification, high crime slows down gentrification.

Better public schools? Good suburban- and faltering urban public schools were often mentioned reason as for suburbanization. During the study period, several reforms aimed at improving urban schools, such as charter schools and vouchers, were introduced. If urban public schools improved as a result, this could be a reason for the city to regain attractiveness. While school quality is an important local amenity, two reasons make us question its role in driving gentrification.

First, through peer effects, parent involvement, and local school financing, school quality is highly endogenous to the local demographics, thus leaving school quality to a large extent outside of the control of "remedial" school policies [Rothstein, 2006]. This does not mean that city public schools have not improved, simply that to the extent they have, gentrification may be the driver.

Second, as mentioned, children are a dwindling demographic in the central city. In 1980, the 6-18 age group made up about 9 percent of the population in the city core, by 2010 that number was close to 6 (Table 1). In other words, the childlessness that characterized early gentrifiers (e.g. yuppies or gays [Black et al., 2002]) remains, perhaps unsurprising considering children's effect on the space-to-earnings ratio).

Population growth While the US and the cities in our sample grew substantially over the study period, not all of our cities grew. In fact, 10 of the 27 cities shrank, losing on average 29% of their population since 1970. Detroit and Saint Louis led the pack, losing almost half its population (Appendix Table A7).[35]

Cities that lost population are of particular interest. As mentioned, growth in the number of rich households may be one reason prices have risen in the central city. For cities that lost half of their populations, this factor is likely to be absent.

Another reason to look at cities by population growth is that real estate is durable and we want to make sure that our results are not driven by price declines as demand rescinds.

Overall, the cities in our sample gained population and congestion may have increased as a result. Since congestion makes commuting longer and less predictable, growing congestion rather than longer work hours may be the reason suburbs have fallen out of fashion. The cities

[35]The reason for using 1970 as the start year rather than 1980 is that we view 1970 as the start year of the processes driving gentrification. Since we obtain similar results in the two groups, we surmise that the exact grouping is not important.

that shrank are instructive in this regard since it is reasonable to believe congestion to have eased or at least not increased in these cities.

Lastly, the potential role of foreign buyers is a reason to look at cities that shrank. Their uncertain economic future and predominant location in the rust belt suggest limited presence of foreign buyers.

Table 6 columns 7 and 8 show the results from splitting the sample by population growth (negative or positive). We see that the results are quite similar in the two subsamples, corroborating the price picture seen in Appendix Figures A1 and A2. The inclusion of distance-year fixed effects (Panel B) affect the magnitude of the coefficient estimates but not the qualitative results.

That is, even for cities that shrank, we find that prices near the CBD have done relatively well. The difference seems to be a matter of levels, population loss resulting in lower overall price pressure. For cities that grew, the demand shock propagated *throughout* the city, but was substantially stronger in the 0-3 mile core.

3.3.2 IV

In the previous set of regressions, we have seen that the inclusion of city-year, city-distance and distance-year fixed effects, while important, leaves little residual variation in the demand shock for our IV strategy. In the presence of weak instruments, IV results could be as severely biased as OLS results. Therefore, we will consider results with city-year and city distance fixed, along with the full set of fixed effects, keeping in mind that the last specification is quite demanding.

Table 7 presents the results. In column 1, FT is instrumented by $(d1_d, d2_d, d3_d) \times Z_{jt}$. We see a statistically significant positive effect, similar in magnitude to the OLS results. However, the positive effect becomes statistically insignificant once distance-year fixed effects are included (column 2). In this specification, the effects of the demand shock on FT mainly holds in the first distance ring (see Table 3, column 4), and the test for weak instruments also points in this direction (the test statistics indicate that the 2SLS bias could be as large as 30%). This is perhaps unsurprising given that the fixed effects absorb much of the variation in our instruments.

Therefore, in column 3 we use only $d1_d \times Z_{jt}$ as instrument (which we know has a strong first stage, Table 3, column 5). The IV estimate is now twice as large as the OLS estimate.

In columns 4 and 5 the effect of FT is allowed to vary with distance, as housing demand close to the CBD likely has a stronger effect on prices (from less elastic supply). We again see a positive effect that attenuates away from the CBD. Once the full set of fixed effects are included, the main effect on FT is slightly reduced, but the linear interacting remains negative

and statistically significant, in contrast to the OLS results.

We note that whenever the full set of fixed effects are included and weak instruments are of concern (columns 2 and 5), the IV results line up in magnitude with the OLS results. However, when instruments are strong (columns 3 and 4) the IV results are larger in magnitude.

Conditional on the necessary IV assumptions being fulfilled, we find that the causal effect of FT on housing prices is larger than the effect estimated by OLS. New construction is a candidate reason why OLS would underestimate the causal effect of FT. While the bulk of residential construction is still greenfield development, since the early 1990s there has been a marked shift towards the city core [Thomas, 2009, 2010, Ramsey, 2012].[36]

The specification in column 3 includes the full set of fixed effects and has a strong first stage. However, it suffers from not allowing the effect of FT on prices to vary with distance from the CBD, a concern since housing supply elasticities vary systematically with distance from the CBD. This limitation is illustrated in Figure 9 which shows the predicted price changes. We see that the model does well close to the CBD but over-predicts price changes away from the CBD.

To allow FT to have a distance differential effect on prices, we revert to using three instruments, $(d1_d, d2_d, d3_d) \times Z_{jt}$. We favor the specification with only metro-year and metro-distance fixed effects (column 4, Table 7) because inclusion of distance-year fixed effects renders the first stage weak (columns 5, Table 7); Figure 10 shows the implied price changes and we see that they track the observed changes closely.

4 Discussion

Suburbanization dominated the US urban landscape for most of the 20th Century. As the century drew to a close, however, it was clear that change was underway. Today, gentrification has grown out of its erstwhile niche status to headline a broad-based revival of the central city.

The driving factor, we have proposed, can be found in the longer hours worked by high-income households. Suburban living offers space against a commute. With little non-market time, the enjoyment of the former may be reduced, while the latter becomes more onerous.

Longer hours by the skilled as the mechanism behind gentrification raises the question why hours and wages have moved in the same direction for the skilled. Higher wages in isolation could have a positive, negative, or neutral effect on hours worked. One reason may lie in the nature of high returns to skill. Long hours may be a form of skill building, and thus more attractive in an environment of high returns to skill. Evidence of hours as training can be seen

[36] We present the IV results for the alternative specifications and samples of Table 6 in the Appendix.

in the bundling of hours and wages. High returns to skill, in turn, are commonly linked to the on-going IT revolution and agglomeration economies which have combined to favor areas with a skilled labor force 1970s, viz. San Jose. These observations motivate our use of a Bartik-type demand shifter, its being particularly apt at capturing exogenous variation in long hours by the skilled.

The ideas collated in this paper hew closely to the canonical models of land and time use. The employment of a Bartik demand shifter also follows a well-beaten path. To the best of our knowledge, however, we are the first to emphasize dwindling non-market time of high-income households as *the* force behind gentrification.

We have focused on individual labor supply and lumped men and women together. In the preliminary analysis we looked at men and women separately but did not find gender differential results, possibly because the expansion of hours have been qualitatively similar for skilled men and women. A counter factual worth considering is whether gentrification would have become the dominant real estate trend absent the decline of the housewife.

The trends studied in this paper are not unique to the US, but their manifestations may be context specific. For instance, while a number of European cities have seen similar price pressure in the central city, the starting points have been different since Europe never had US-style suburbanization. The urban real estate in place at the time of suburbanization certainly played a role [Hohenberg and Lees, 1995, Brueckner et al., 1999]. Less discussed, however, is the greater purchasing power of the US middle class. Circa 1850, US overtook Western Europe in terms of GDP per capita, and by 1950 the latter was a mere half of the former.[37] In other words, the limited purchasing power of the European middle class post WWII may have contributed to their choosing existing city-core real estate rather than greenfield construction, leaving the latter for low-cost or public housing.

Gentrification is about price growth and changes to the housing stock, not population growth. The suburbs will likely continue to be the destination of choice, if not aspiration, for the vast majority of people. However, our paper has pointed to gentrification having its roots in the labor market for the skilled, rather than greater demand for other central city amenities, or being part of a cultural pendulum swing. The distinction may seem academic, but the implications are vastly different. If gentrification is the result of local amenities, country clubs and high-end restaurants, perhaps with local farmer tie ins, could anchor suburbs, and there is no telling where the next hot spot might be. If gentrification is simply the rejection of decades of suburbanization, the suburbs have a bright future ahead of them once this cycle has run its course. By contrast, if little non-market time is behind gentrification, two clear

[37]The Maddison-Project `http://www.ggdc.net/maddison/maddison-project/home.htm`

policy recommendations emerge: (i) relax hight/density restrictions on buildings in the city center; and (ii) invest in mass transit that is time efficient, frequent, and impervious to road congestion, viz. rail transit.

References

Mark Aguiar and Erik Hurst. *The Increase in Leisure Inequality 1965-2005*. The AEI Press, 2009.

David Albouy and Bert Lue. Driving to opportunity: Local rents, wages, commuting, and sub-metropolitan quality of life. *Journal of Urban Economics*, 89:74 – 92, 2015. ISSN 0094-1190.

David H. Autor and David Dorn. The growth of low-skill service jobs and the polarization of the US labor market. *The American Economic Review*, 103(5), August 2013.

David H. Autor, Lawrence F. Katz, and Melissa S. Kearney. Trends in U.S. wage inequality: Revising the revisionists. *The Review of Economics and Statistics*, 90(2):300–323, May 2008.

Timothy J. Bartik. *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute for Employment Research, Kalamazoo, MI, 1991.

C. F. Baum. Enhanced routines for instrumental variables/generalized method of moments estimation and testing. *Stata Journal*, 7(4):465–506(42), 2007. URL `http://www.stata-journal.com/article.html?article=st0030_3`.

Nathaniel Baum-Snow and Daniel Hartley. Causes and consequences of central neighborhood change, 1970-2010. May 2016. URL `https://docs.google.com/viewer?a=v&pid=sites&srcid=ZGVmYXVsdGRvbWFpbnxiYXVtc25vd3xneDo1NzE3ZTcxNTk0ZjcxM2M`.

Paul Beaudry, Mark Doms, and Ethan Lewis. Should the Personal Computer Be Considered a Technological Revolution? Evidence from US Metropolitan Areas. *Journal of Political Economy*, 118(5):988–1036, October 2010.

Christopher R. Berry and Edward L. Glaeser. The divergence of human capital levels across cities. *Papers in Regional Science*, 84(3):407 – 444, 2005. ISSN 10568190.

Dan Black, Gary Gates, Seth Sanders, and Lowell Taylor. Why do gay men live in San Francisco? *Journal of Urban Economics*, 51(1):54 – 76, 2002.

Dan A. Black, Natalia A. Kolesnikova, and Lowell J. Taylor. Why do so few women work in New York (and so many in Minneapolis)? labor supply of married women across U.S. cities. *Journal of Urban Economics*, 79(January):59–71, 2014.

Willem R. Boterman, Lia Karsten, and Sako Musterd. Gentrifiers settling down? patterns and trends of residential location of middle-class families in Amsterdam. *Housing Studies*, 25(5): 693 – 714, 2010.

Jan K. Brueckner and Stuart S. Rosenthal. Gentrification and neighborhood housing cycles: will America's future downtowns be rich? *Review of Economics and Statistics*, 91(4):725–743, November 2009.

Jan K. Brueckner, Jacques-Francois Thisse, and Yves Zenou. Why is central Paris rich and downtown Detroit poor?: An amenity-based theory. *European Economic Review*, 43(1):91 – 107, 1999.

Juliet Carpenter and Loretta Lees. Gentrification in New York, London and Paris: an international comparison. *International Journal of Urban and Regional Research*, 19(2):286, 1995.

Janice Compton and Robert A. Pollak. Why are power couples increasingly concentrated in large metropolitan areas? *Journal of Labor Economics*, 25(3):pp. 475–512, 2007.

Dora Costa and Matthew E. Kahn. Power couples: Changes in the locational choice of the college educated, 1940-1990. *Quarterly Journal of Economics*, 115(4):1287–1315, November 2000.

Victor Couture and Jessie Handbury. Urban revival in America, 2000 to 2010. November 2015. URL `http://faculty.haas.berkeley.edu/couture/download/Couture_Handbury_Revival.pdf`.

Judie B. Cullen and Steven D. Levitt. Crime, urban flight, and the consequences for cities. *Review of Economics and Statistics*, 81(2):159–169, May 1999.

Morris A. Davis and Jonathan Heathcote. The price and quantity of residential land in the United States. *Journal of Monetary Economics*, 54(8):2595 – 2620, 2007.

Rebecca Diamond. The determinants and welfare implications of US workers' diverging location choices by skill: 1980-2000. *American Economic Review*, 2016.

Riccardo DiCecio, Kristie M. Engemann, Michael T. Owyang, and Christopher H. Wheeler. Changing trends in the labor force: A survey. *Federal Reserve Bank of St. Louis Review*, 90 (1), January/February 2008.

John J. III Donohue and Steven D. Levitt. The impact of legalized abortion on crime. *Quarterly Journal of Economics*, 116(2):379–420, May 2001.

Lena Edlund, Cecilia Machado, and Maria Sviatschi. Bright minds, big rent: Gentrification and the rising returns to skill. Working Paper 21729, National Bureau of Economic Research, November 2015.

Ingrid Gould Ellen and Katherine O'Regan. Crime and urban flight revisited: The effect of the 1990s drop in crime on cities. *Journal of Urban Economics*, 68(3):247 – 259, 2010.

Ingrid Gould Ellen and Katherine M. O'Regan. How low income neighborhoods change: Entry, exit, and enhancement. *Regional Science and Urban Economics*, 41(2):89–97, MAR 2011.

Giovanni Favara and Jean Imbs. Credit supply and the price of housing. *American Economic Review*, 105(3):958–92, 2015.

Christopher Foote and Christopher Goetz. The impact of legalized abortion on crime: Comment. *Quarterly Journal of Economics*, 123(1), February 2008.

Leigh Gallagher. *The End of the Suburbs: Where the American Dream Is Moving.* Portfolio/Penguin, 2014.

Edward L. Glaeser, Jed Kolko, and Albert Saiz. Consumer city. *Journal of Economic Geography*, (1):27–50, 2001.

Edward L. Glaeser, Matthew E. Kahn, and Jordan Rappaport. Why do the poor live in cities? the role of public transportation. *Journal of Urban Economics*, 63(1):1 – 24, 2008.

Edward L. Glaeser, Joshua D. Gottlieb, and Joseph Gyourko. Can cheap credit explain the housing boom. Technical report, Harvard University and Wharton, 2010.

Claudia Goldin. A grand gender convergence: Its last chapter. *American Economic Review*, 104(4):1091–1119, 2014.

Jeremy Greenwood and Boyan Jovanovic. The information-technology revolution and the stock market. *The American Economic Review*, 89(2):pp. 116–122, 1999.

Veronica Guerrieri, Daniel Hartley, and Erik Hurst. Endogenous gentrifcation and housing price dynamics. *Journal of Public Economics*, 100:45–60, 2013.

Joseph Gyourko, Christopher Mayer, and Todd Sinai. Superstar cities. *American Economic Journal: Economic Policy*, 5(4):167–99, 2013.

Bradley T. Heim. The incredible shrinking elasticities: Married female labor supply, 1978-2002. *The Journal of Human Resources*, 42(4):pp. 881–918, 2007.

Andrew C. Helms. Understanding gentrification: an empirical analysis of the determinants of urban housing renovation. *Journal of Urban Economics*, 54(3):474 – 498, 2003.

Bart Hobijn and Boyan Jovanovic. The information-technology revolution and the stock market: Evidence. *The American Economic Review*, 91(5):pp. 1203–1220, 2001.

Paul M. Hohenberg and Lynn Hollen Lees. *The making of urban Europe, 1000-1994*. Harvard University Press, Cambridge, MA, 1995.

Jackelyn Hwang and Jeffrey Lin. What have we learned about the causes of recent gentrification? *Cityscape*, forthcoming.

William R. Johnson. House prices and female labor force participation. University of Virginia, January 2012.

Chinhui Juhn and Kevin M. Murphy. Wage inequality and family labor supply. *Journal of Labor Economics*, 15(1):72–97, January 1997.

Chinhui Juhn, Kevin M. Murphy, and Brooks Pierce. Wage inequality and the rise in returns to skill. *Journal of Political Economy*, 101(3):410–442, June 1993.

Lawrence F. Katz and Kevin M. Murphy. Changes in the relative wages, 1963-87: supply and demand factors. *Quarterly Journal of Economics*, 107(1):35–78, February 1992.

Peter J. Kuhn and Fernando A. Lozano. The expanding workweek? understanding trends in long work hours among U.S. men, 1979-2004. Discussion Paper 1924, IZA, 2006.

Janice F. Madden and Michelle J. White. Spatial implications of increases in the female labor force: A theoretical and empirical synthesis. *Land Economics*, 56(4):432–446, Nov. 1980.

Janice Fanning Madden. Consequences of the growth of the two-earner family urban land use and the growth in two-earner households. *American Economic Review*, 70(2):191–197, 1980.

Ellen McGrattan and Richard Rogerson. Changes in the distribution of family hours worked since 1950. Research Department Staff Report 397, Federal Reserve Bank of Minneapolis, 2008.

Brian McKenzie and Melanie Rapino. Commuting in the United States, 2009. American Community Survey Reports ACS-15, Census Bureau, Washington, DC, 2011.

Terra McKinnish, Randall Walsh, and T. Kirk White. Who gentrifies low-income neighborhoods? *Journal of Urban Economics*, 67(2):180 – 193, 2010.

Peter Mieszkowski and Edwin S. Mills. The causes of metropolitan suburbanization. *Journal of Economic Perspectives*, 7(3):135–147, Sum 1993.

Jacob Mincer. Family migration decisions. *Journal of Political Economy*, 86(5):749–773, October 1978.

Enrico Moretti. Real wage inequality. *American Economic Journal-Applied Economics*, 5(1): 65–103, Jan 2013.

National Association of Realtors. Home buyer and seller generational trends. Technical report, 2014. http://www.realtor.org/sites/default/files/reports/2014/2014-home-buyer-and-seller-generational-trends-report-full.pdf.

Walter Oi. Residential location and labor supply. *Journal of Political Economy*, 84(4):S221–S238, 1976.

Francois Ortalo-Magne and Andrea Prat. On the political economy of urban growth: Homeownership versus affordability. *American Economic Journal: Microeconomics*, 6(1):154–81, 2014.

Kevin Ramsey. Residential construction trends in Americas metropolitan regions. Technical report, U.S. Environmental Protection Agency, 2012. URL `https://www.epa.gov/sites/production/files/2014-03/documents/residential_construction_trends.pdf`.

Jordan Rappaport. Monocentric city redux. Research Working Paper 14-09, The Federal Reserve Bank of Kansas City, 2014.

Jordan Rappaport. Productivity, congested commuting, and metro size. Research Working Paper 16-03, The Federal Reserve Bank of Kansas City, 2016.

Jessica Wolpaw Reyes. Environmental policy as social policy? The impact of childhood lead exposure on crime. *B E Journal of Economic Analysis & Policy*, 7(1), 2007.

Jesse M. Rothstein. Good principals or good peers? parental valuation of school characteristics, Tiebout equilibrium, and the incentive effects of competition among jurisdictions. *American Economic Review*, 96(4):1333–1350, September 2006.

Christine R. Schwartz. Earnings inequality and the changing association between spouses earnings. *American Journal of Sociology*, 115(5):pp. 1524–1557, 2010.

James H. Stock and Motohiro Yogo. Testing for weak instruments in linear iv regression. In Donald W. K. Andrews and James H. Stock, editors, *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*. Cambridge University Press, 2005. http://dx.doi.org/10.1017/CBO9780511614491.006.

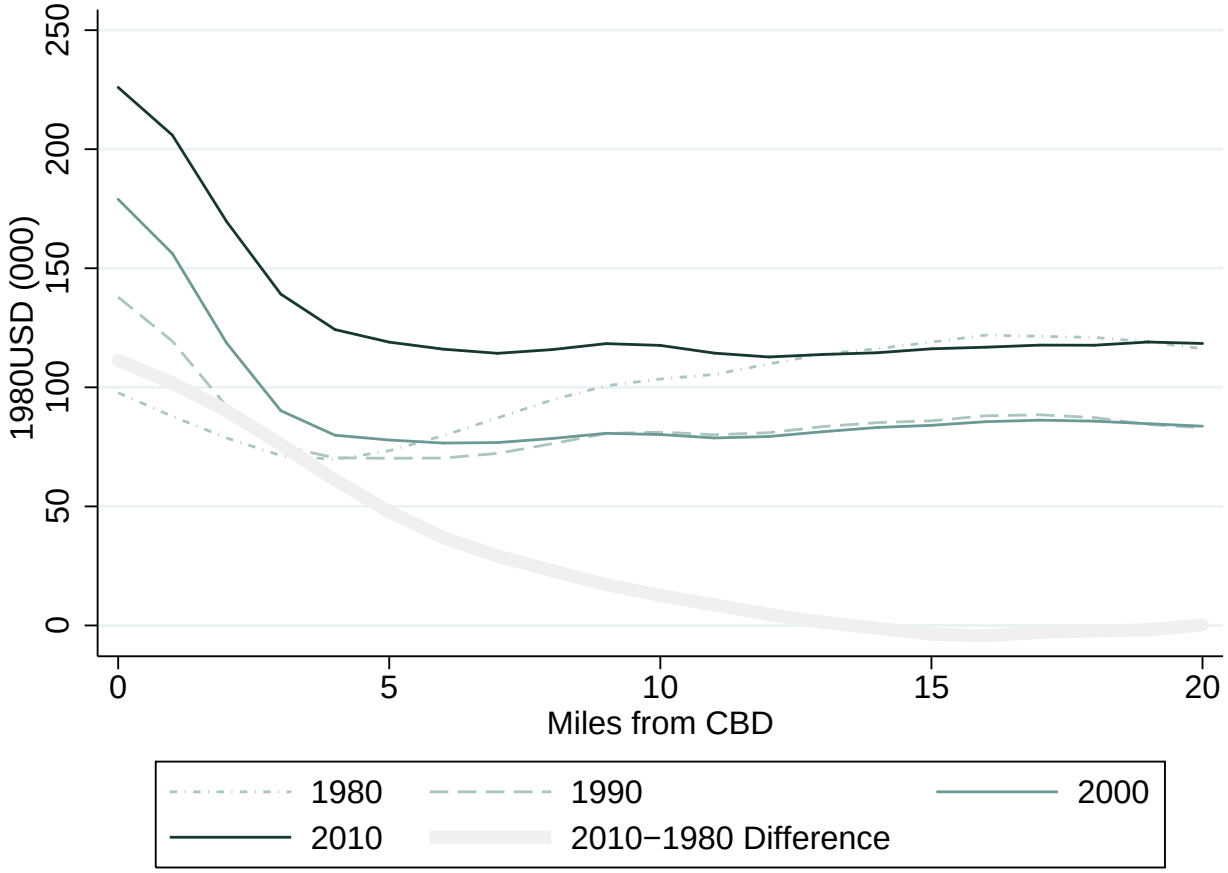
John Thomas. Residential construction trends in Americas metropolitan regions. Technical report, U.S. Environmental Protection Agency, 2009. URL http://www.epa.gov/smartgrowth/construction_trends.htm.

John Thomas. Residential construction trends in Americas metropolitan regions. Technical report, U.S. Environmental Protection Agency, 2010. URL https://www.epa.gov/sites/production/files/2014-03/documents/metro_res_const_trends_10.pdf.

Michelle J. White. A model of residential location choice and commuting by men and women workers. *Journal of Regional Science*, 17(1):41–51, 1977.

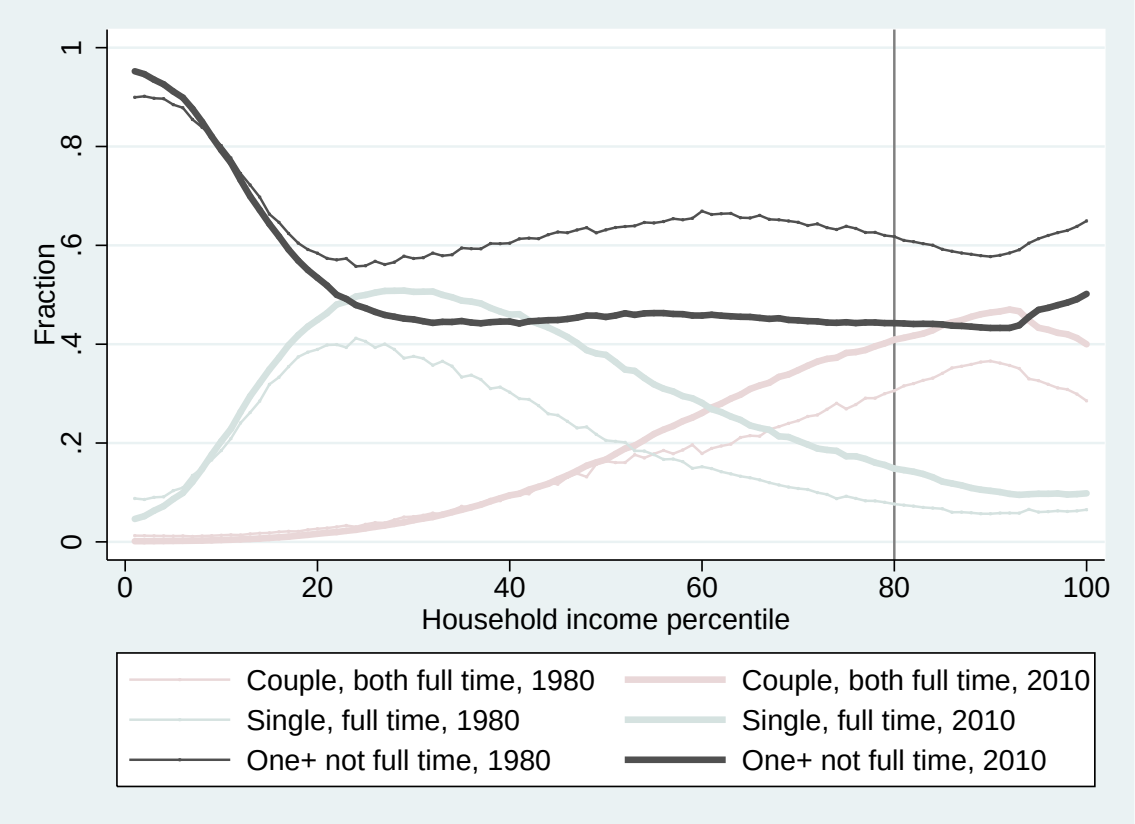
Graphs

Figure 1: Home prices by distance from the CBD



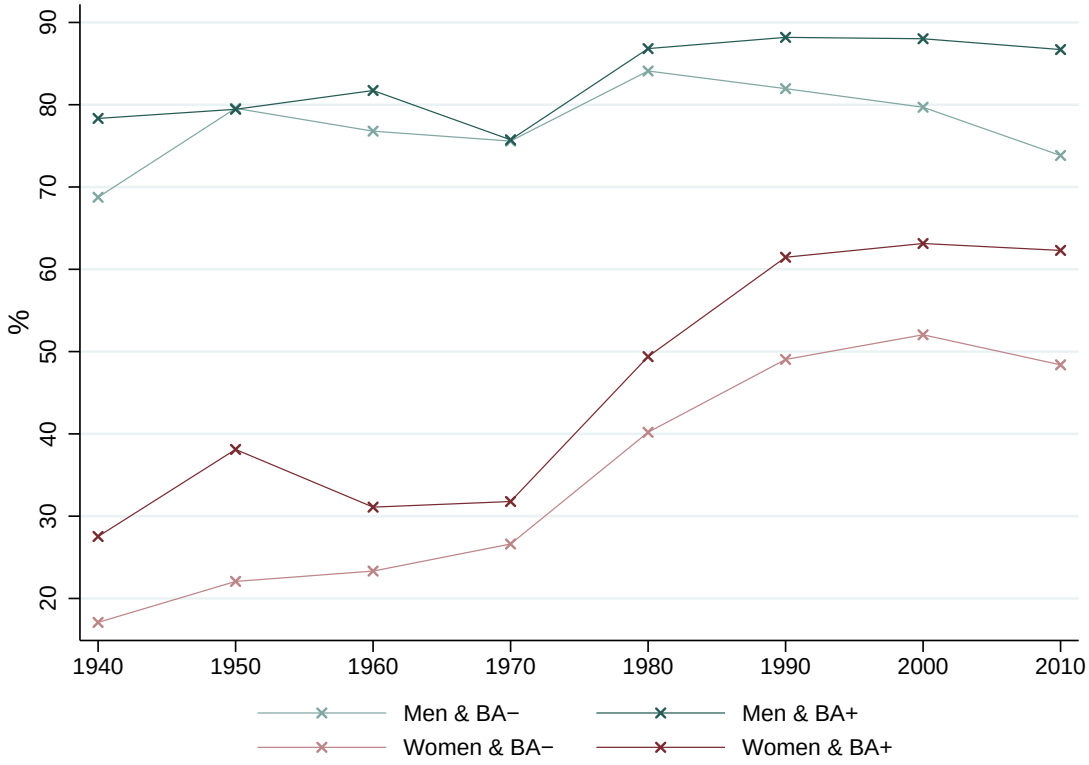
Notes: The figure shows the median price (1980$) for owner-occupied, 2-3 bedroom, one-family homes in our top-20 US cities, by distance from the CBD. See the Appendix for further details on variable and sample construction, notably Table A8. 20 miles includes 20-35 miles.
Source: Decennial censuses and the American Community Survey, restricted use data.

Figure 2: Households by householders full-time status, 1980 and 2010



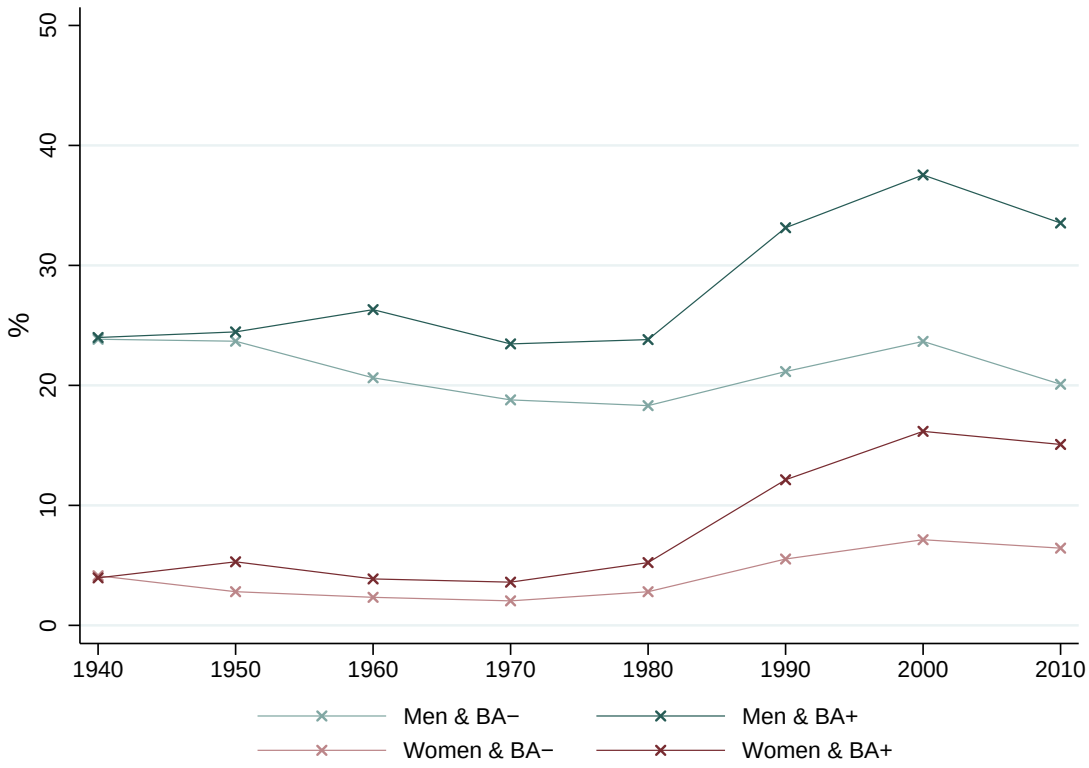
The black lines show the fraction of households at each percentile in the household income distribution that has at least one of its householders (head or spouse) who does not work full time (40 hours or more per week, 40 or more hours per year); it is 1 minus the fraction single (blue line) and couple (red line) households where head and spouse (if applicable) work full time. Household income is computed as the sum of total personal income for head and spouse. The universe is all households in which either head or spouse is between ages 25 and 55 and not in group quarters.
Source: IPUMS.

Figure 3: % Full time, 40+h/week, by sex and education



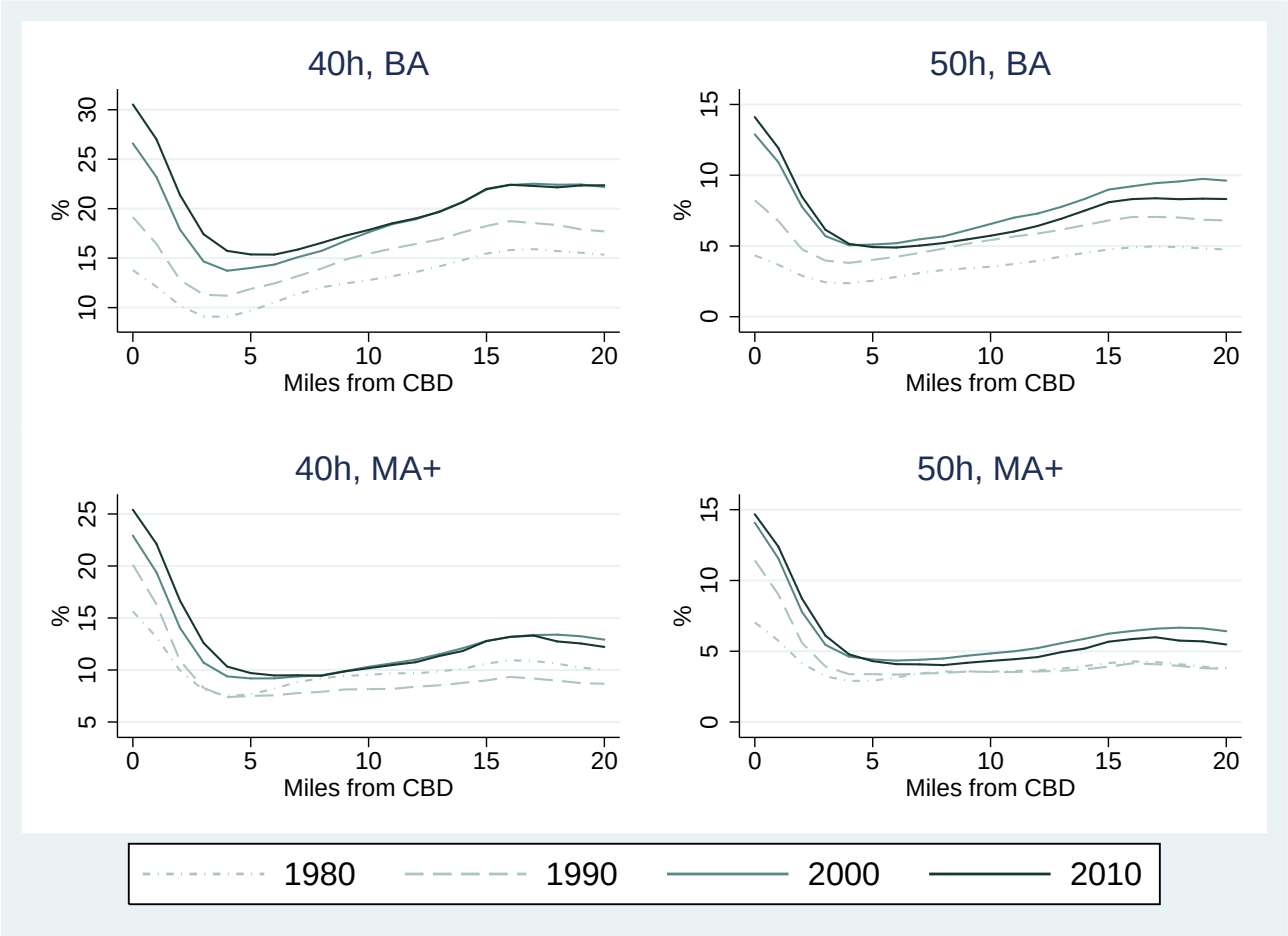
Note: Ages 25-55.
Source: Decennial censuses, integrated public use micro data series (IPUMS).

Figure 4: % Full time, 50+h/week, by sex and education



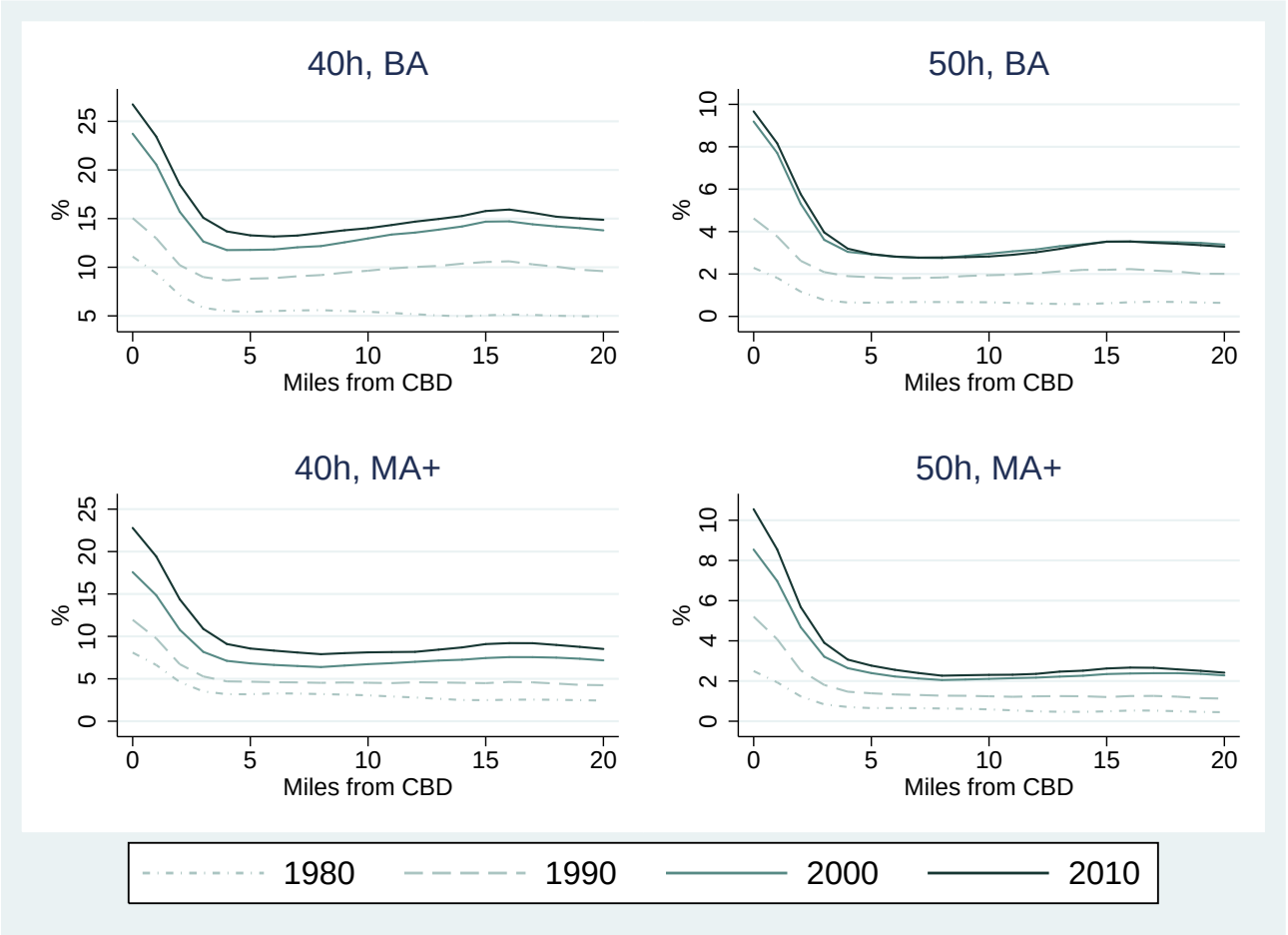
Note: Ages 25-55.
Source: Decennial censuses, integrated public use micro data series (IPUMS).

Figure 5: % Full time and skilled by distance from the CBD, men 25-55



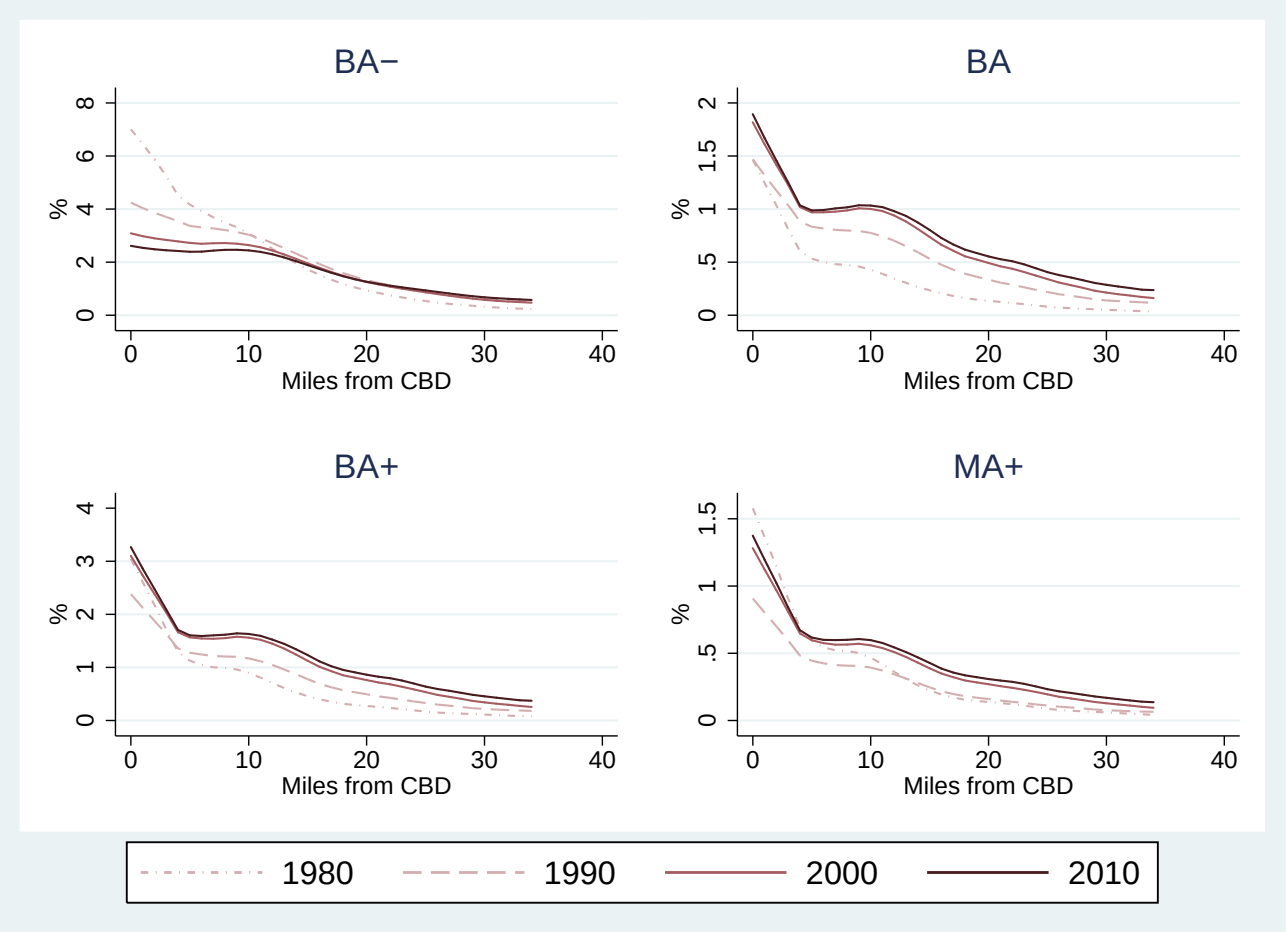
Universe: Men 25-55 residing $x = 0, 1, ...18, 19, 20 - 35$ miles from the CBD. Source: Decennial censuses and the American Community Survey, restricted use data.

Figure 6: % Full time and skilled by distance from the CBD, women 25-55



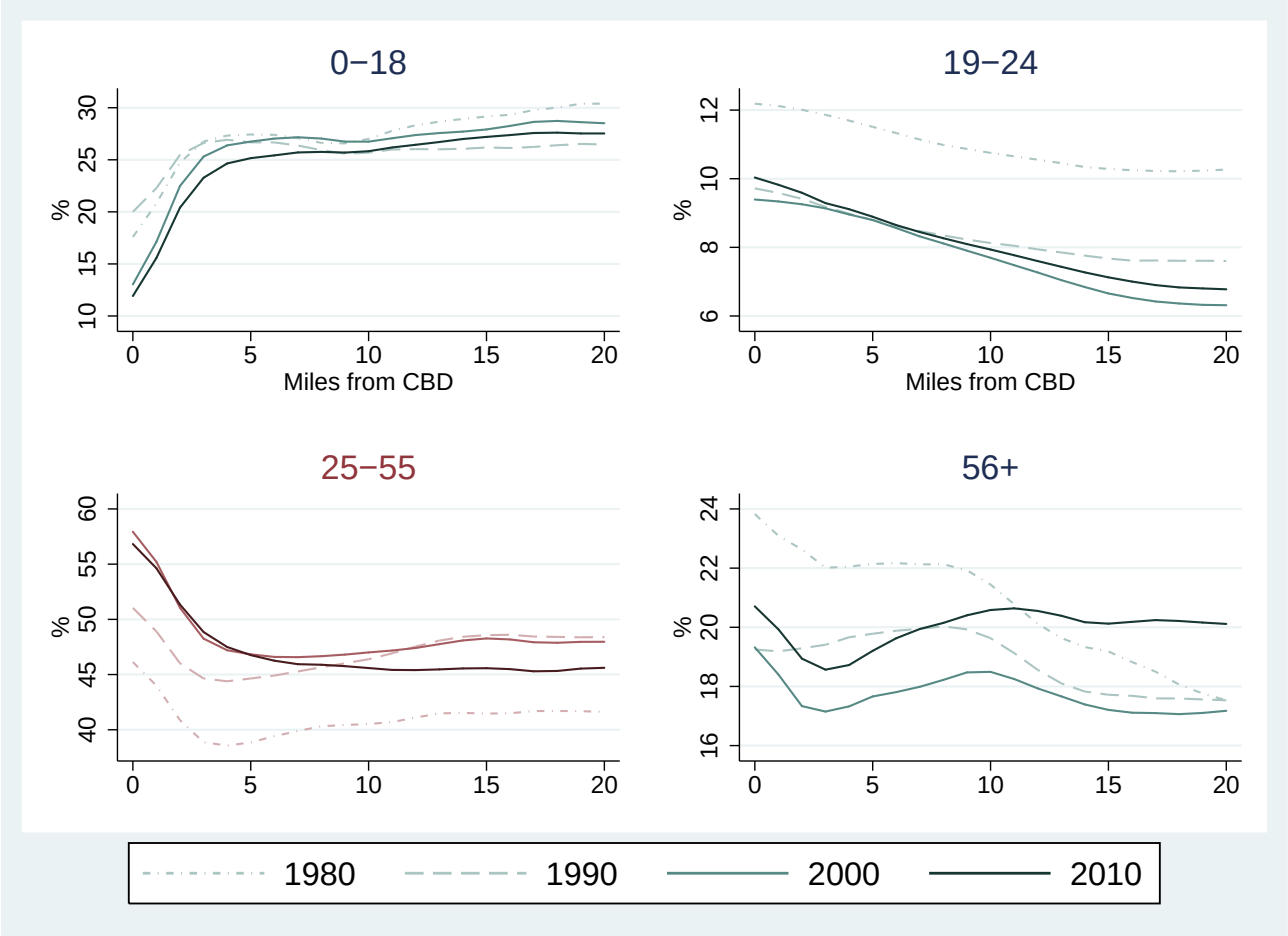
Universe: Women 25-55 residing $x = 0, 1, ...18, 19, 20 - 35$ miles from the CBD. Source: Decennial censuses and the American Community Survey, restricted use data.

Figure 7: Skill and distance distribution of jobs



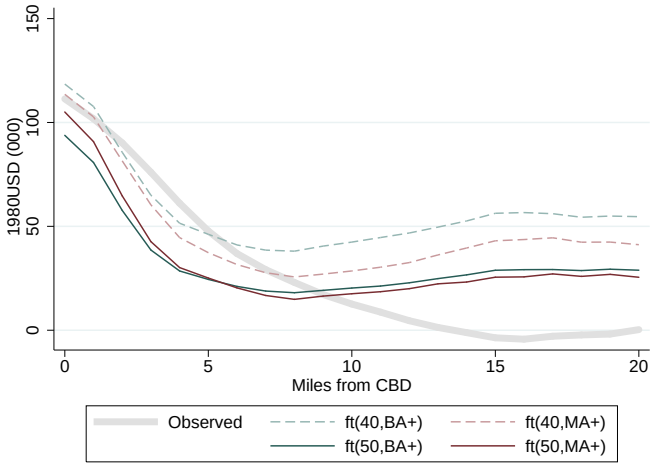
Notes: The universe is the jobs within 35 miles of the CBD in a given year, held by workers 25-55 who work away from home. For instance, the bottom left panel says that in 2000, 3 percent of jobs were located in the center (0-1 mile radius) and were held by a worker with four year college or more.
The skill level of the job is given by the education level of the person who held the job.
Source: Decennial censuses and the American Community Survey, restricted use data.

Figure 8: Age distribution by residence location



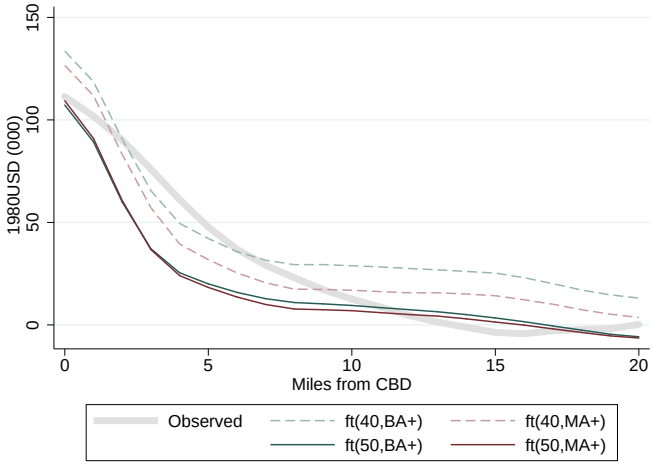
Universe: Residents residing $x = 0, 1, ...18, 19, 20 - 35$ miles from the CBD.
Source: Decennial censuses and the American Community Survey, restricted use data.

Figure 9: Predicted v. Actual Price increase, '000 1980$



This figure graphs the predicted price change from Table 7, column 3.

Figure 10: Predicted v. Actual Price increase, '000 1980$



This figure graphs the predicted price change from Table 7, column 4.

Tables

Unless otherwise specified, the analysis is based on the decennial censuses and the American Community Survey, restricted use data. Throughout, sample sizes have been rounded to nearest 1000 for disclosure reasons. The repeated cross section has 65 thousand tract-years (53 thousand excluding New York); and the panel data set has 48 thousand tract-years.

Table 1: Summary Statistics

		(1)	(2)	(3)	(4)	(5)
		Distance to the CBD (miles)				
Variable	Year	$d1$=[0,3]	$d2$=(3,10]	$d3$=(10,20]	$d4$=(20,35)	[0,35)
House price	1980	66.64	79.16	108.23	114.06	92.5
('000 1980$)	2010	154.77	115.07	119.22	120.5	120.54
	Δ	88.13	35.91	10.99	6.44	28.04
FT(40,BA+)	1980	13.78	12.92	15.19	15.05	14.05
	2010	35.13	23.15	27.22	28.29	26.62
	Δ	21.35	10.23	12.03	13.24	12.57
FT(40,MA+)	1980	6.37	5.42	6.12	5.64	5.77
	2010	15.5	8.92	10.1	10.35	10.13
	Δ	9.13	3.5	3.98	4.71	4.36
FT(50,BA+)	1980	4.03	3.43	4.3	4.44	3.92
	2010	14.32	7.25	8.81	9.69	8.86
	Δ	10.29	3.82	4.51	5.25	4.94
FT(50,MA+)	1980	2.29	1.82	2.06	1.93	1.96
	2010	7.28	3.4	3.8	4.02	3.95
	Δ	4.99	1.58	1.74	2.09	1.99
Income	1980	10.53	12.06	14.25	14.3	12.94
('000 1980$)	2010	17.98	13.99	17.22	18.77	16.53
	Δ	7.45	1.93	2.97	4.47	3.59
White	1980	48.99	62.43	75.61	85.73	68.87
	2010	44.97	38.81	53.91	66.37	51.06
	Δ	-4.02	-23.62	-21.7	-19.36	-17.81
Black	1980	32.89	23.61	13.41	6.78	18.7
	2010	25	27.22	15.43	7.49	18.25
	Δ	-7.89	3.61	2.02	0.71	-0.45

Table 1: Summary Statistics, continued

		(1)	(2)	(3)	(4)	(5)
		Distance to the CBD (miles)				
Variable	Year	$d1$=[0,3]	$d2$=(3,10]	$d3$=(10,20]	$d4$=(20,35)	[0,35)
Married	1980	46.29	62.67	72.22	77.35	66.32
	2010	36.13	48.25	60.72	66.59	56.14
	Δ	-10.16	-14.42	-11.5	-10.76	-10.18
Ages:						
0-5	1980	8.03	7.85	7.51	8.23	7.82
	2010	6.99	8.32	8.04	8.12	8.08
	Δ	-1.04	0.47	0.53	-0.11	0.26
6-12	1980	9.04	9.56	10.09	11.02	9.9
	2010	6.58	9.02	9.63	10.12	9.33
	Δ	-2.46	-0.54	-0.46	-0.9	-0.57
13-18	1980	8.97	9.66	10.62	11.35	10.16
	2010	6.05	8.12	8.64	8.83	8.33
	Δ	-2.92	-1.54	-1.98	-2.52	-1.83
19-24	1980	12.45	11.21	10.41	10.46	10.97
	2010	10.31	8.58	7.36	6.77	7.84
	Δ	-2.14	-2.63	-3.05	-3.69	-3.13
25-55	1980	39	38.7	40.52	40.81	39.63
	2010	50.45	45.53	44.64	44.63	45.34
	Δ	11.45	6.83	4.12	3.82	5.71
56-65	1980	9.74	10.86	10.67	9.21	10.42
	2010	9.67	9.88	10.7	10.88	10.39
	Δ	-0.07	-0.98	0.03	1.67	-0.03
66-	1980	12.76	12.16	10.18	8.93	11.09
	2010	9.95	10.55	10.99	10.64	10.68
	Δ	-2.81	-1.61	0.81	1.71	-0.41

Note: percent unless otherwise indicated. Income is total personal income. Source: Decennial censuses and the American Community Survey, restricted use data.

Table 2: Effects of Bartik-type demand shock on housing values

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Dependent Variable: House price ('000) 1980$								
Z	522.0*	425.7							
	[278.0]	[264.1]							
$Z\times$									
$d1$		588.9***	580.7***	633.1***	777.4***	561.7**			
		[168.6]	[169.1]	[123.1]	[264.1]	[259.7]			
$d2$		234.5***	249.7***	285.2***	235.6				
		[52.21]	[49.95]	[54.47]	[144.8]				
$d3$		88.25***	98.22***	153.8***	319.5*				
		[18.25]	[18.49]	[41.01]	[173.0]				
$Z\times$									
$dist$							-50.30***	-47.79***	-42.65***
							[13.92]	[7.843]	[9.668]
$dist^2$							1.053***	0.928***	0.819***
							[0.307]	[0.156]	[0.150]
R^2	0.317	0.333	0.362	0.400	0.401	0.401	0.353	0.401	0.402
Fixed effects:									
City	✓	✓							
Year	✓	✓							
City-Year			✓	✓	✓	✓	✓	✓	✓
City-Distance				✓	✓	✓		✓	✓
Distance-Year					✓	✓			✓

Regressions 1-3 include distance controls $d1, d2, d3$; regressions 7-9 include linear and square continuous distance controls ($dist, dist^2$).
Z is the Bartik demand shifter, see Equation 4, also Table A8. $d1, d2$, and $d3$ indicate distance intervals 0-3, 3-10, 10-20 miles from the CBD, respectively.
The unit of observation is a tract and each regression has 65-thousand tract-years.
Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55).
Source: Decennial censuses and the American Community Survey, restricted use data.

Table 3: Effects of Bartik-type demand shock on full-time skilled workers

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable:				
	A. $FT(40, BA+)$				
Z	11.56				
	[13.91]				
$Z\times$					
$d1$		97.27***	33.10**	120.6*	129.4***
		[12.58]	[13.43]	[59.46]	[39.51]
$d2$		16.94	-31.30***	24.02	
		[14.58]	[8.369]	[44.49]	
$d3$		6.977	-16.11*	-54.85*	
		[16.93]	[8.549]	[29.14]	
R^2	0.185	0.203	0.318	0.320	0.320
	B. $FT(40, MA+)$				
Z	11.73**				
	[4.520]				
$Z\times$					
$d1$		52.17***	14.22***	52.46**	57.52***
		[6.621]	[4.024]	[21.24]	[16.07]
$d2$		12.34	-15.66***	6.307	
		[10.97]	[3.892]	[17.57]	
$d3$		8.080	-7.354	-22.21	
		[10.78]	[4.823]	[15.33]	
R^2	0.167	0.185	0.283	0.285	0.284
Fixed effects:					
city	✓				
year	✓				
city-year		✓	✓	✓	✓
city-distance			✓	✓	✓
distance-year				✓	✓

Regressions 1-2 include distance controls $d1, d2, d3$.
Z is the Bartik demand shifter, see Equation 4, also Table A8. $d1, d2$, and $d3$ indicate distance 0-3, 3-10, 10-20 miles from the CBD, respectively.
The unit of observation is a tract and each regression has 65-thousand tract-years.
Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55).
Source: Decennial censuses and the American Community Survey, restricted use data.

Table 4: Effects of Bartik-type demand shock on other demographics

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent Variable:					
	%					('000) 1980$
	$FT(40, BA-)$	$BA+$	Black	White	Married	Income
$Z\times$						
$d1$	-0.62	85.72	-92.67	268.2***	103.9***	63.70*
	[0.376]	[55.53]	[67.05]	[49.59]	[24.01]	[34.16]
$d2$	0.029	14.69	-52.56	141.7**	70.97**	9.523
	[0.314]	[59.31]	[52.06]	[58.50]	[27.07]	[14.24]
$d3$	0.516*	-76.31*	-17.5	63.12	51.39**	-5.304
	[0.284]	[39.51]	[51.28]	[58.72]	[18.62]	[13.84]
R^2	0.261	0.291	0.279	0.444	0.477	0.295

All regressions include city-year, city-distance and distance-year fixed effects.
Z is the Bartik demand shifter, see Equation 4, also Table A8. $d1, d2$, and $d3$ indicate distance intervals 0-3, 3-10, 10-20 miles from the CBD, respectively.
The unit of observation is a tract and each regression has 65-thousand tract-years.
Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55). Source: Decennial censuses and the American Community Survey, restricted use data.

Table 5: The relationship between full-time skilled workers and housing prices

	(1)	(2)	(3)	(4)
	Dependent Variable: House price ('000) 1980$			
	A. $FT(40, BA+)$			
FT	1.829***	1.831***	2.037***	1.838***
	[0.167]	[0.167]	[0.164]	[0.147]
$FT \times dist$			-0.0170**	-0.000484
			[0.00686]	[0.00501]
R^2	0.475	0.482	0.475	0.482
	B. $FT(40, MA+)$			
FT	3.502***	3.507***	3.635***	3.334***
	[0.340]	[0.342]	[0.335]	[0.309]
$FT \times dist$			-0.0115	0.0144
			[0.0163]	[0.0130]
R^2	0.475	0.482	0.475	0.482
Fixed effects:				
City-Year	✓	✓	✓	✓
City-Distance	✓	✓	✓	✓
Distance-Year		✓		✓

Regressions 3-4 include continuous distance control, *dist*.
The unit of observation is a tract and each regression has 65-thousand tract-years.
Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55). Source: Decennial censuses and the American Community Survey, restricted use data.

Table 6: Effects of Bartik-type demand shock on housing values – robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent Variable: House price ('000) 1980$							
	Sample:							
		Cross Section:						
		Years:		Crime decline:			Population:	
	Panel	1980/1990	2000/2010	High	Low	not NYC	Shrunk	Grew
	Panel A. City-year and City-distance FE							
$Z\times$								
$d1$	655.5***	752.3***	1164.2***	728.6***	511.2***	493.3***	596.9***	661.3***
	[169.4]	[143.5]	[179.5]	[187.8]	[92.62]	[62.16]	[88.75]	[190.4]
$d2$	299.7***	449.8***	531.4**	285.2***	290.3**	266.7***	352.6***	241.0***
	[60.29]	[112.4]	[228.5]	[43.39]	[114.6]	[71.79]	[100.7]	[66.12]
$d3$	125.2*	298.7***	175.8**	99.77**	241.4***	150.1**	239.9**	102.8**
	[61.03]	[107.0]	[79.86]	[37.41]	[70.42]	[57.38]	[77.41]	[41.01]
R^2	0.591	0.188	0.613	0.356	0.346	0.408	0.241	0.436
	Panel B. City-year, City-distance, and Distance-year FE							
$Z\times$								
$d1$	856.2**	533.2	2464.4***	998.3	565.7***	591.8***	622.2***	1190.0**
	[375.1]	[454.2]	[635.1]	[618.4]	[108.0]	[111.3]	[148.9]	[514.0]
$d2$	271	-146.3	1489.6	359.5	244.8**	225.1*	264.2	552.3**
	[194.1]	[444.8]	[1000.5]	[231.5]	[112.0]	[125.3]	[149.0]	[223.9]
$d3$	431.6*	456.6	280.2	473.6***	398.3***	325.8*	474.6***	354.8**
	[251]	[552.0]	[180.3]	[120.5]	[117.2]	[170.1]	[146.7]	[151.3]
R^2	0.562	0.188	0.613	0.356	0.348	0.409	0.244	0.436

Column 1 includes tract fixed effects.
Z is the Bartik demand shifter, see Equation 4, also Table A8. $d1, d2$, and $d3$ indicate distance 0-3, 3-10, 10-20 miles from the CBD, respectively.
The unit of observation is a tract and sample sizes are as follows: column 1: 48 thousand; columns 2-5, 7-8: about 65/2 thousand; column 6: 53 thousand tract-years.
Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55). Source: Decennial censuses and the American Community Survey, restricted use data.

Table 7: Effects of full-time skilled workers on housing values – IV results

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: House price ('000) 1980$				
	A. $FT(40, BA+)$				
FT	2.007*	1.71	4.341***	5.026***	3.429**
	[1.201]	[1.363]	[1.215]	[1.049]	[1.508]
$FT \times dist$				-0.194***	-0.401**
				[0.0185]	[0.166]
K-P LM test (p)	0.00615	0.0353	0.00704	0.0024	0.03
C-D $Wald$ stat.	$77.35^{(b5,s10)}$	$32.45^{(b5,s10)}$	$53.9^{(b10)}$	$76.86^{(b5,s10)}$	$12.49^{(b5,s15)}$
K-P $Wald$ stat.	$20^{(b5,s10)}$	$6.007^{(b30)}$	$10.82^{(b10)}$	$18.99^{(b5,s10)}$	3.023
Overid. test (p)	0.0456	0.105	.	0.228	0.29
	B. $FT(40, MA+)$				
FT	3.212	4.595	9.765***	11.11***	5.943
	[2.605]	[3.232]	[2.759]	[2.408]	[4.049]
$FT \times dist$				-0.499***	-0.666**
				[0.0541]	[0.320]
K-P LM test (p)	0.00438	0.0233	0.0053	0.0018	0.0589
C-D $Wald$ stat.	$63.02^{(b5,s10)}$	$20.08^{(b5,s10)}$	$38.88^{(b10)}$	$62.22^{(b5,s10)}$	$9.431^{(b5,s15)}$
K-P $Wald$ stat.	$30.01^{(b5,s10)}$	4.68	$12.92^{(b10)}$	$20.62^{(b5,s10)}$	1.54
Overid. test (p)	0.0589	0.149	.	0.195	0.219
Fixed effects:					
City-Year	✓	✓	✓	✓	✓
City-Distance	✓	✓	✓	✓	✓
Distance-Year		✓	✓		✓
Instruments:					
$Z \times (d1, d2, d3)$	✓	✓		✓	✓
$Z \times d1$			✓		

Notes on separate page.

Notes to Table 7.

All specifications include metro-year metro-distance and distance-year fixed effects.

Regressions 4-5 include continuous distance control, *dist*.

Z is the Bartik demand shifter, see Equation 4, also Table A8. *dist* is distance in miles from the CBD, $d1, d2$, and $d3$ indicate distance intervals 0-3, 3-10, 10-20 miles from the CBD, respectively.

The unit of observation is a tract and each regression has 65-thousand tract-years.

Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55). Source: Decennial censuses and the American Community Survey, restricted use data.

K-P LM test (p) corresponds to the p-value of the Kleinbergen-Paap LM test. The null hypothesis is that the structural equation is under identified (i.e., the rank condition fails).

C-D $Wald$ stat and K-P $Wald$ statistics are the Cragg-Donald and Kleibergen-Paap Wald statistics for testing weak identification. In both cases, the critical values are the Stock and Yogo [2005] critical values initially tabulated for the C-D $Wald$ stat, see Table 8. We follow Baum [2007] and also apply the Stock and Yogo critical values to the K-P $Wald$ stat (critical values for the K-P $Wald$ stat do not exist). In each case, we specify whether the test statistics rejects the null hypothesis (at the 5% level) that the bias of of the IV estimates exceeds the OLS bias by 5, 10, 20 and 30 percent (b5, b10, b20, b30), and whether test statistics rejects the null hypothesis (at the 5% level) that the size of the test is greater than 10, 15, 20 and 25 percent (s20, s15, s20, s25). C-D $Wald$ stat and K-P $Wald$ statistics reduce to the the standard non-robust and heteroscedasticity robust first-stage F-statistics, respectively. The critical values for the relative bias test cannot be computed for the case of 2 endogenous and 3 instruments and we use the more conservative critical values for the case of 2 endogenous and 4 instruments, see Stock and Yogo [2005] and Table 8.

Overid. test (p) corresponds to the p-value of the test of the overidentifying restrictions. The null hypothesis is that the instruments are valid instruments.

Source: Decennial censuses and the American Community Survey, restricted use data.

Table 8: Threshold values for Cragg-Donald and Kleibergen-Paap $Wald$ statistics

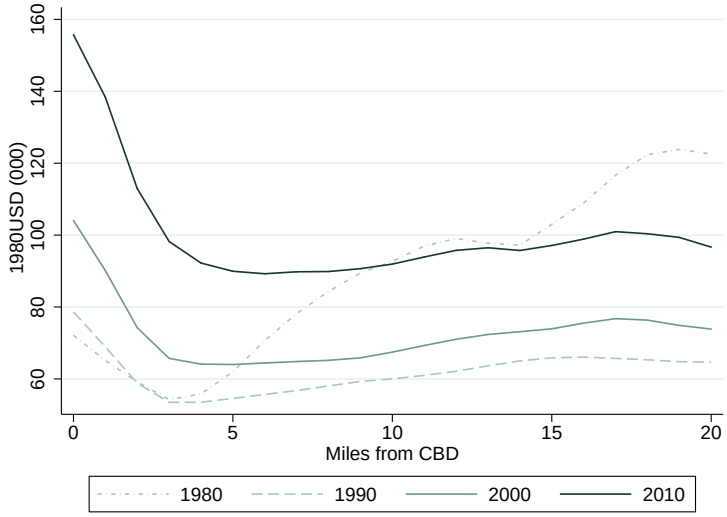
	(1)	(2)
Number of variables:		
Endogenous	1	2
Exogenous	3	4^a
Stock-Yogo 2SLS Bias		
0.05	13.91	11.04
0.10	9.08	7.56
0.20	6.46	5.57
0.30	5.39	4.73
Stock-Yogo 2SLS Size		
0.10	22.30	13.43
0.15	12.83	8.18
0.20	9.54	6.40
0.25	7.80	5.45

^{a} – used for our case of 2 endogenous and 3 exogenous (more conservative).
Source: Stock and Yogo [2005, table 5.1 and 5.2].

Appendix

Figures

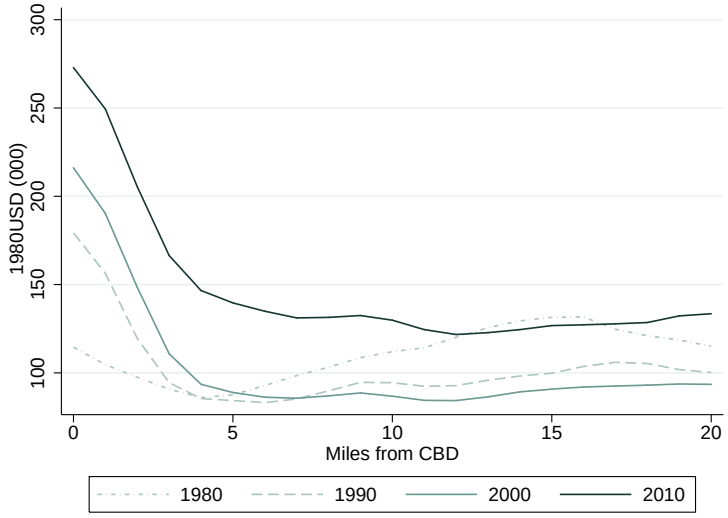
Figure A1: Home prices by distance from the CBD, cities that lost population



Notes: The figure shows the median price (1980$) for owner-occupied, 2-3 bedroom, one-family homes in our top-20 US cities, by distance from the CBD. See the Appendix for further details on variable and sample construction, especially Table A7. 20 miles includes 20-35 miles.

Source: Decennial censuses and the American Community Survey, restricted use data.

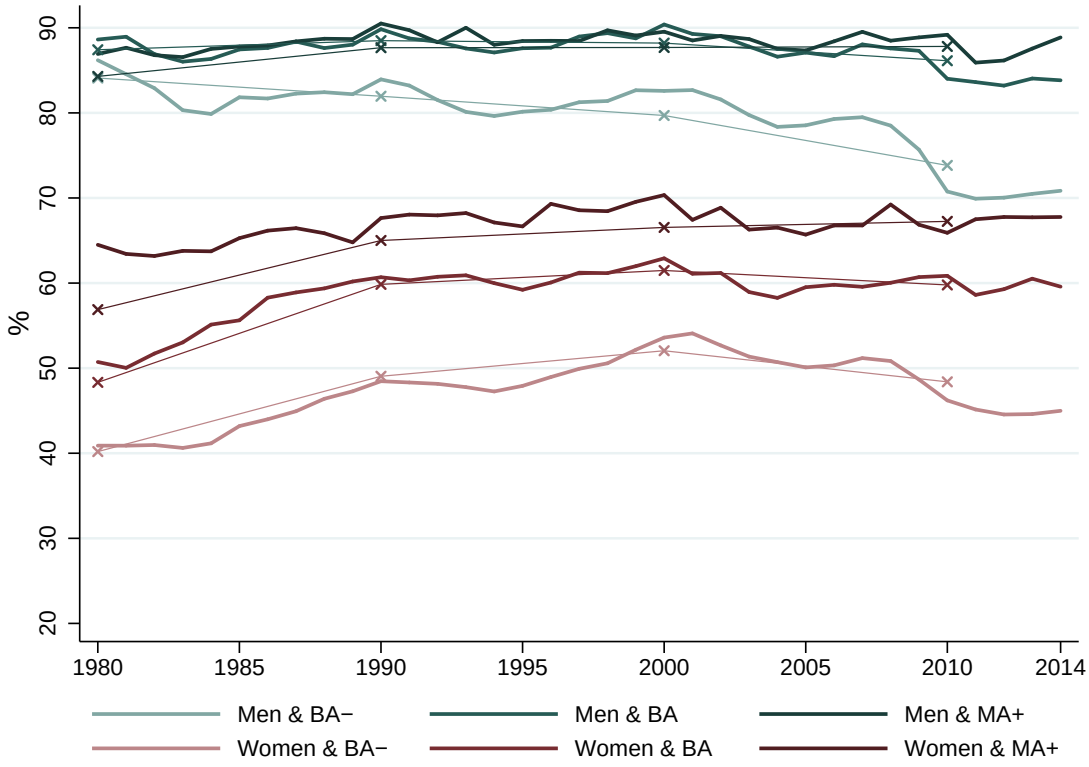
Figure A2: Home prices by distance from the CBD, cities that gained population



Notes: The figure shows the median price (1980$) for owner-occupied, 2-3 bedroom, one-family homes in our top-20 US cities, by distance from the CBD. See the Appendix for further details on variable and sample construction, especially Table A7. 20 miles includes 20-35 miles.

Source: Decennial censuses and the American Community Survey, restricted use data.

Figure A3: Full time (40+h/week)



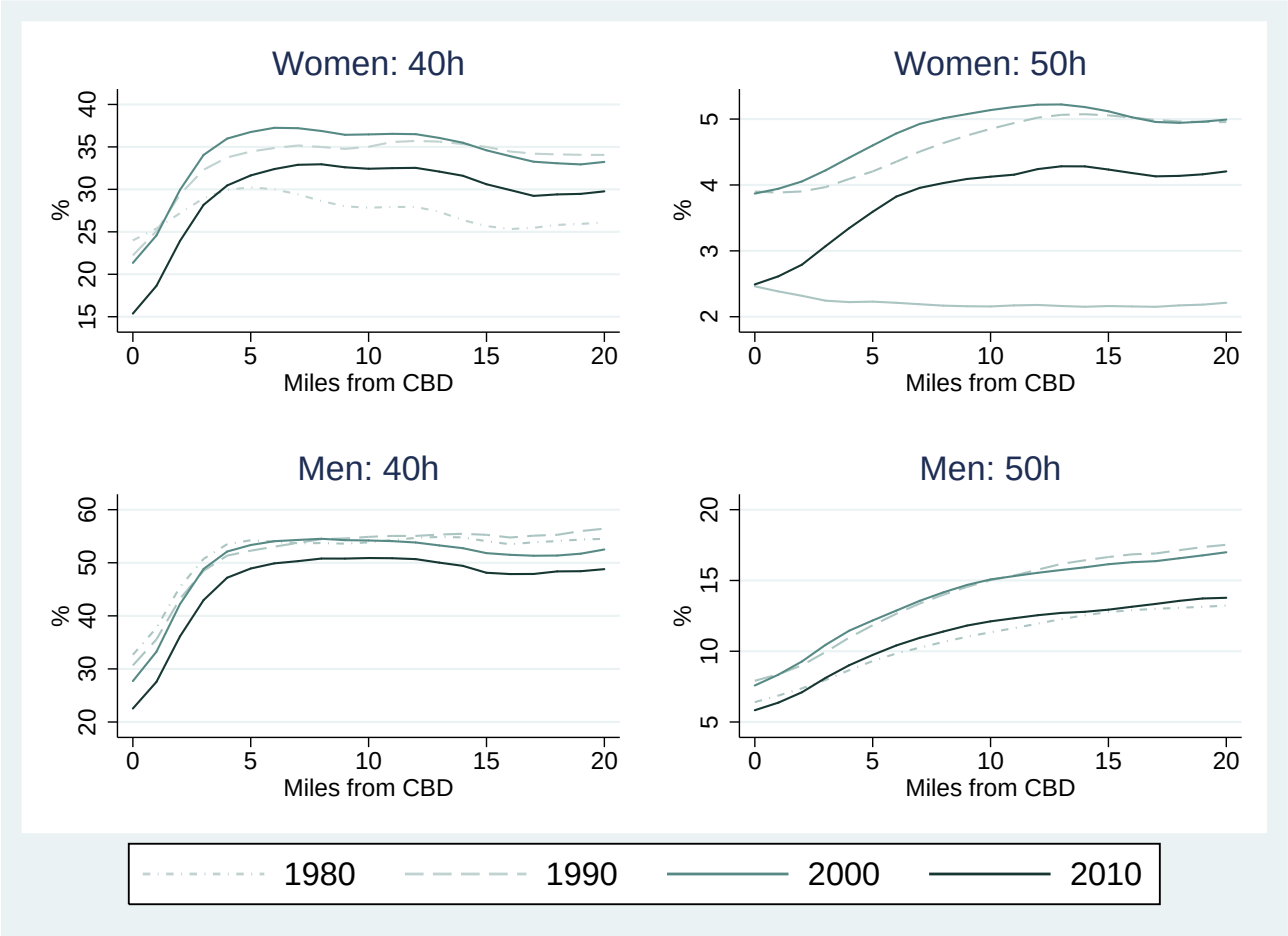
Note: Men and women ages 25-55.
Sources: Decennial data: Decennial censuses, integrated public use micro data series (IPUMS). Annual data: The Current Population Surveys (CPS).

Figure A4: Full time (50+h/week)



Note: Men and women ages 25-55.
Sources: Decennial data: Decennial censuses, integrated public use micro data series (IPUMS). Annual data: The Current Population Surveys (CPS).

Figure A5: % adults 25-55, unskilled and full time, by distance from the CBD



Notes: 40h(50h) denotes working 40(50) hours or more per week.
Source: Decennial censuses and the American Community Survey, restricted use data.

Tables

Table A1: The Effects of Demand Shock on Housing Values ('000) 1980$, logged

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Dependent Variable: House price ('000) 1980$, Logged								
Z	1.995	1.212							
	[1.585]	[1.622]							
$Z\times$									
$d1$		4.988***	5.031***	3.618***	-1.148	-2.363**			
		[1.157]	[1.172]	[0.581]	[1.438]	[1.009]			
$d2$		2.442***	2.588***	1.798***	1.386				
		[0.605]	[0.602]	[0.529]	[1.501]				
$d3$		0.626***	0.682***	0.720***	1.725*				
		[0.188]	[0.182]	[0.259]	[0.860]				
$Z\times$									
$dist$							-0.532***	-0.335***	-0.246***
							[0.122]	[0.0506]	[0.0833]
$dist^2$							0.0116***	0.00703***	0.00526**
							[0.00270]	[0.00119]	[0.00144]
R^2	0.479	0.498	0.528	0.591	0.592	0.592	0.528	0.592	0.593
Fixed effects:									
City	✓	✓							
Year	✓	✓							
City-Year			✓	✓	✓	✓	✓	✓	✓
City-Distance				✓	✓	✓		✓	✓
Distance-Year					✓	✓			✓

Notes, see Table 2.

Table A2: Effects of Bartik-type demand shock on full-time skilled workers

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable:				
	$FT(50, BA+)$				
Z	11.73				
	[8.483]				
$Z\times$					
$d1$		54.97***	35.51**	103.3**	95.50**
		[11.44]	[14.73]	[44.69]	[42.18]
$d2$		8.181	-14.00***	25.65*	
		[7.638]	[3.863]	[14.84]	
$d3$		2.720	-8.558**	-9.807	
		[7.520]	[4.119]	[13.34]	
R^2	0.145	0.167	0.294	0.296	0.296
	$FT(50, MA+)$				
Z	8.160***				
	[1.981]				
$d1$		29.22***	16.79***	43.35**	40.07**
		[4.266]	[5.363]	[17.56]	[17.98]
$d2$		6.175	-6.297***	10.76	
		[5.229]	[2.014]	[7.386]	
$d3$		3.229	-3.270	-4.061	
		[4.468]	[2.224]	[6.950]	
R^2	0.128	0.147	0.253	0.255	0.254
Fixed effects:					
city	✓				
year	✓				
city-year		✓	✓	✓	✓
city-distance			✓	✓	✓
distance-year				✓	✓

Notes, see Table 3.

Table A3: The relationship between full-time skilled workers and housing prices

	(1)	(2)	(3)	(4)
	$FT(50, BA+)$			
FT	4.003***	3.995***	4.320***	3.964***
	[0.371]	[0.371]	[0.325]	[0.322]
$FT \times dist$			-0.0260**	0.00227
			[0.0124]	[0.0118]
R^2	0.488	0.494	0.488	0.494
	$FT(50, MA+)$			
FT	6.732***	6.708***	6.740***	6.210***
	[0.651]	[0.654]	[0.625]	[0.587]
$FT \times dist$			-0.00124	0.0422
			[0.0320]	[0.0283]
R^2	0.480	0.486	0.480	0.486
Fixed effects:				
City-Year	✓	✓	✓	✓
City-Distance	✓	✓	✓	✓
Distance-Year		✓		✓

Notes, see Table 5.

Table A4: Effects of full-time skilled workers on housing values – IV results

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: House price ('000) 1980$				
	$FT(50, BA+)$				
FT	6.017***	4.328***	5.882***	7.067***	6.887***
	[0.889]	[1.325]	[1.047]	[0.801]	[2.613]
$FT \times dist$				-0.401***	-1.483
				[0.0430]	[1.136]
K-P LM test (p)	0.0163	0.0174	0.00521	0.00677	0.177
C-D $Wald$ stat.	169.4	53	121.8	147.9	2.511
K-P $Wald$ stat.	7.615	4.755	5.17	8.768	1.982
Overid. test (p)	0.0299	0.405	.	0.328	0.91
	$FT(50, MA+)$				
FT	13.15***	10.34***	14.02***	15.28***	10.37**
	[1.973]	[3.234]	[2.421]	[1.787]	[5.090]
$FT \times dist$				-0.915***	-2.401
				[0.107]	[2.362]
K-P LM test (p)	0.0164	0.0142	0.00437	0.0025	0.199
C-D $Wald$ stat.	111.2	28.59	65.86	96.59	2.211
K-P $Wald$ stat.	10.05	4.506	5.008	11.44	0.907
Overid. test (p)	0.0406	0.378	.	0.336	0.463
Fixed effects:					
City-Year	✓	✓	✓	✓	✓
City-Distance	✓	✓	✓	✓	✓
Distance-Year		✓	✓		✓
Instruments:					
$Z \times (d1, d2, d3)$	✓	✓		✓	✓
$Z \times d1$			✓		

Notes, see Table 7.

Table A5: Effects of full-time skilled workers on housing values – IV results, robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent Variable: House price ('000) 1980$							
	Sample:							
		Cross Section:						
		Years:		Crime decline:			Population:	
	Panel	1980/1990	2000/2010	High	Low	not NYC	Shrunk	Grew
	$FT(40, BA+)$							
FT	4.132***	5.267	6.339**	3.767***	3.477*	3.402***	4.266***	4.473***
	[1.407]	[3.768]	[2.599]	[0.793]	[1.943]	[0.671]	[1.379]	[1.647]
K-P LM test (p)	0.0037	0.0444	0.146	0.128	0.0337	0.0496	0.172	0.017
C-D $Wald$ stat.	436.7	6.407	6.727	13.13	40.27	36.78	11.37	43.02
K-P LM test (p)	10.23	12.68	9.509	22.64	22.84	13.03	23.63	31.48
	$FT(40, MA+)$							
FT	9.750***	8.132	8.083***	7.075***	8.704*	7.570***	8.175***	11.39***
	[2.687]	[5.489]	[2.142]	[2.258]	[4.796]	[1.612]	[2.787]	[4.037]
K-P LM test (p)	0.0028	0.0692	0.0943	0.092	0.0318	0.0349	0.183	0.0112
C-D $Wald$ stat.	240.4	10.27	14.87	13.88	22.91	27.74	10.03	26.71
K-P LM test (p)	8.121	6.568	27.51	11.08	16.78	18.36	27.4	15.52
	$FT(50, BA+)$							
FT	5.306***	5.203*	15.67**	7.124***	4.487***	5.326***	7.154***	5.756***
	[1.182]	[2.977]	[7.663]	[1.345]	[1.572]	[1.197]	[1.622]	[1.145]
K-P LM test (p)	0.00436	0.0212	0.132	0.141	0.0193	0.0542	0.114	0.00663
C-D $Wald$ stat.	720.1	35.13	4.338	16.2	94.37	65.25	17.15	106.3
K-P LM test (p)	5.354	8.844	6.511	22.75	6.095	8.368	13.83	9.061
	$FT(50, MA+)$							
FT	12.69***	8.600*	19.72***	15.77***	10.92***	12.72***	15.65***	14.55***
	[2.379]	[4.846]	[5.624]	[4.675]	[3.563]	[2.897]	[3.431]	[2.749]
K-P LM test (p)	0.00548	0.0278	0.0862	0.0993	0.0183	0.0399	0.102	0.00676
C-D $Wald$ stat.	347.8	37.95	8.411	10.14	48.88	35.46	9.747	56.16
K-P LM test (p)	4.376	7.545	20.72	11.22	5.489	8.8	12.24	7.126

The instrument is $Z \times d1$, corresponding to column 3, Table 7.
For cities in columns 4 and 5, see Appendix Table A6.
For cities in columns 7 and 8, see Appendix Table A7.
All specifications include metro-year metro-distance and distance-year fixed effects.
Z is the Bartik demand shifter, see Equation 4, also Table A8. $d1, d2$, and $d3$ indicate distance 0-3, 3-10, 10-20 miles from the CBD, respectively.
The unit of observation is a tract and sample sizes are as follows: column 1: 48 thousand; columns 2-5, 7-8: about 65/2 thousand; column 6: 53 thousand tract-years.
Significance levels: *** – 0.01; ** – 0.05; * – 0.1. Standard errors are clustered at the city level; tracts are weighted by population size (25-55). Source: Decennial censuses and the American Community Survey, restricted use data.
See Table 7 for further notes.

Table A6: Change in Violent Crime 1986-2012

High Crime Reduction Cities	
New Orleans	-.64
New York City	-.60
Boston	-.55
Los Angeles County	-.54
Detroit	-.42
Baltimore City	-.41
Dallas	-.41
San Francisco	-.39
Low Crime Reduction Cities	
St. Louis	-.34
Jacksonville	-.28
City Of Fort Worth	-.28
Washington Metropolitan	-.26
San Diego County Sheriff	-.18
El Paso County Sheriff	-.18
Cleveland	-.17
San Jose	.02
Philadelphia	.10
Columbus*	.18
Memphis	.18
Phoenix	.25
San Antonio	.28
Houston	.31
Austin	.44
Milwaukee	.91
Indianapolis	1.15
Chicago	n.a.
Charlotte	n.a

* The entry for Columbus is for the period 1986-2011.

Violent crimes include murder, rape, robbery, aggravated assault. For a detailed description, see `http://www.ucrdatatool.gov/offenses.cfm`.

Source: Uniform Crime Report `http://www.ucrdatatool.gov/Search/Crime/Local/TrendsInOneVarLarge.cfm`

Table A7: Population Change

City	Population 1970	Population 2010	Change	% Change
		Shrunk		
Detroit	1,511,482	713,777	-797,705	-53
St. Louis	622,236	319,294	-302,942	-49
Cleveland	750,903	396,815	-354,088	-47
New Orleans	593,471	343,829	-249,642	-42
Baltimore	905,759	620,961	-284,798	-31
Philadelphia	1,948,609	1,526,006	-422,603	-22
DC	756,510	601,723	-154,787	-20
Chicago	3,366,957	2,695,598	-671,359	-20
Milwaukee	717,099	594,833	-122,266	-17
Boston	641,071	617,594	-23,477	-4
Sum	11,814,097	8,430,430	-3,383,667	-29
		Grew		
New York	7,894,862	8,175,133	280,271	4
Memphis	623,530	646,889	23,359	4
Indianapolis	744,624	820,445	75,821	10
SF	715,674	805,235	89,561	13
LA	2,816,061	3,792,621	976,560	35
Columbus	539,677	787,033	247,356	46
Jacksonville	528,865	821,784	292,919	55
Dallas-FW	1,237,877	1,939,012	701,135	57
Houston	1,232,802	2,100,263	867,461	70
San Diego	696,769	1,307,402	610,633	88
El Paso	322,261	649,121	326,860	101
San Antonio	654,153	1,327,407	673,254	103
San Jose	445,779	945,942	500,163	112
Phoenix	581,562	1,445,632	864,070	149
Charlotte	241,178	731,424	490,246	203
Austin	251,808	790,390	538,582	214
Sum	19,527,482	27,085,733	7,558,251	39

Source: US Census.

Bartik

We group industries into 41 categories using the 1990 industry classification available in IPUMS (see IPUMS website or Edlund et al. [2015]. In the preliminary analysis we also used a more aggregate grouping into seven industries, the results were very similar, but we favor the more disaggregate grouping because of the reduction in within-group industry heterogeneity.

Table A8: Bartik, Base Year 1970

City	1980	1990	2000	2010	Central Business District (CBD)
Austin	0.169	0.259	0.320	0.350	Texas Capitol
Baltimore	0.110	0.169	0.206	0.223	W. Lexington × Park Ave.
Boston	0.144	0.226	0.277	0.298	South Station
Charlotte	0.102	0.162	0.192	0.206	Charlotte Convention Center
Chicago	0.110	0.175	0.214	0.229	LaSalle × W. Congress Parkway
Cleveland	0.102	0.162	0.196	0.212	Soldiers' and Sailors' Monument
Columbus	0.126	0.197	0.242	0.262	E. Long Street × Route 3
Dallas	0.122	0.196	0.232	0.251	Dallas Convention center
Detroit	0.090	0.145	0.171	0.186	Grand Circus Park
El Paso	0.112	0.170	0.211	0.230	El Paso Art Institute
Fort Worth	0.122	0.196	0.232	0.251	Fort Worth Convention Center
Houston	0.137	0.209	0.243	0.266	Houston Center
Indianapolis	0.093	0.147	0.177	0.191	Monument Circle
Jacksonville	0.094	0.148	0.178	0.194	Bank of America Tower
Los Angeles	0.126	0.200	0.242	0.260	Pershing Square
Memphis	0.091	0.144	0.176	0.190	Beale St. × Riverside Drive
Milwaukee	0.102	0.159	0.192	0.207	Milwaukee County Court House
New Orleans	0.112	0.169	0.196	0.216	New Orleans Morial Convention Center
New York	0.136	0.217	0.266	0.286	Rockefeller Center
Philadelphia	0.105	0.165	0.202	0.218	City Hall
Phoenix	0.116	0.183	0.221	0.237	Phoenix Convention Center
Saint Louis	0.103	0.163	0.197	0.213	Federal Reserve
San Antonio	0.089	0.138	0.166	0.180	Tower of the Americas
San Diego	0.125	0.196	0.240	0.259	Horton Plaza
San Francisco	0.172	0.273	0.332	0.362	Transamerica Pyramid
San Jose	0.163	0.253	0.312	0.330	1 Infinity Loop (Apple Inc. headquarters)
Washington D.C.	0.229	0.344	0.426	0.461	White House

Variable construction

PRICE **2-3 Bedroom single-family Home** The decennial censuses and the ACS ask the owner of owner-occupied single-family homes the estimated value of the value of their home. While self-reported assessments of values, we will refer to them as housing prices.

In order to obtain numbers that are close to market prices, we restrict the sample to those who moved in within the last ten years, on the assumption that homes that owners of more recently bought and sold units would be more knowledgeable about price developments than those with longer tenure.

To obtain a price that refers to comparable units while preserving sample size, we focus

on two or three bedroom single-family homes. Housing prices are given in intervals and the bracket values vary across the years. Therefore, we focus on the median bracket value. For tracts where the median bracket is the top code, we assign the dollar value that would be the mean if the top bracket had had the same range as the penultimate bracket. For instance, if the penultimate bracket ranged from 800 thousand to 1 million, and 1 million and above were the top bracket, we would give houses in the top bracket a 1.1 million dollar valuation. We chose this rule because it is conservative and if anything result in an underestimate of the price increases close to the CBD.

We focus on the median house price in the tract. Top coding is our reason for focusing on the median rather than a more selective percentile.

In our main specification, we impute a value that is the same distance from the top-code threshold as the immediately preceding midpoint value. For instance, if the top bracket is 800-1000, and values above 1000 are top coded, we assign the value 1100 to the top code. This method is conservative if the housing price distribution, like the wealth and income distributions, is right skewed.

We also tried alternative top codings, including simply imputing the threshold value. Our qualitative results were not sensitive to these variations.

For the study period 1980-2010, our measure shows a 30 percent increase in constant dollars, a rise largely inline with the Case-Shiller national index's rise of 26 percent. The Case-Shiller national index went from 43.44 to 145.0, and the CPI from 100 to 264, yielding a constant dollar housing price increase of 26 percent ($=\frac{145}{43.44} \times \frac{100}{264} - 1$).

Another benchmark is offered by the Census Bureau's constant price index for new-family homes sold rose from \$72 to \$272 thousand, current dollars. In constant dollars, the increase was 30 percent ($=\frac{261}{76} \times \frac{100}{264} - 1$). `https://www.census.gov/const/www/constpriceindex_excel.html`

We use the Consumer Price Index (CPI) to deflate housing prices. A theoretical concern is that housing accounts for some 40 percent of the CPI and therefore a price index that excluded housing would be preferable, for instance the food price index. Conveniently, the food price index moved very closely with the CPI and on those grounds we use the CPI despite its theoretical shortcomings.

Center point, Central Business District (CBD) For each city we identified the center with the help of Google maps. All cities had a clear central area identifiable from the convergence of roads, the presence of a main railway station, clusters of hotels with the national chain name prefixed by "down town" and a concentration of signature institutions and monuments. We used the thus identified area to designate a city center, with one exception. For San Jose, we placed downtown in Silicon Valley.

The only city with more than one clear center was New York City, where both midtown and downtown can claim that title. We picked the midtown center but locating the center downtown resulted in similar results. Because of this ambiguity, in a robustness test we exclude the New York metropolitan area and results are robust to this exclusion (Table A5, column 3).

Within each center we picked a center point, a salient building or monument, and obtained its latitude and longitude from iTouchMap.com. For instance, for Washington DC, the CBD is given by the White House and for San Jose, Apple Inc. Headquarters. While clearly there are alternative points, but most contenders would be within a mile or two of the points picked, listed in Table A8.

Distance to the CBD (*dist*) From tract shape files, we have generated latitude and longitude for the (population weighted) centroid of the tract, allowing us to calculate the distance between a tract and the CBD. We restrict our sample to tracts that are within 35 miles of the CBD.

In the preliminary analysis, we found our variables of interest to exhibit a pronounced j-shape with respect to distance from the CBD, the minimum located somewhere in the 3-7 mile range. Therefore we grouped distance from the CBD as follows:

d1 0-3 miles

d2 3-10 miles

d3 10-20 miles

d4 20-35 miles, reference category.

FT **Full Time** $FT(h, e)$ denotes the fraction of adults 25-55 who worked more that $h, h = 40, 50$ hours per week and had education $e, e = BA+, MA+$.

Z **Metropolitan Area Demand Shifter for Skilled Workers** See description in Section 2.2

IV.1 Panel data set construction

We use US2010 cross-walk files to create a tract-level panel data set, where for each tract in 2010 we construct its equivalent in previous years using cross-walk files from `http://www.s4.brown.edu/us2010/Researcher/Bridging.htm`. These files provide a mapping of tracts in a census year and 2010, as well as weights. For example, if there were a new tract B in 2010 that was the result of combining blocks from year 2000 tracts B1 and B2, then we would create a year 2000 tract B as a weighted average of B1 and B2. Thus, the ability to include tract fixed effects comes at the cost of data quality, and the problem worsens with the number of years to 2010 (more time allows for more changes to tracts). In order to explore the role of race, we are interested in parsing the sample by initial fraction blacks. Therefore, we restrict the sample to tracts for which we have an observation in 1980 and also remove tracts for which we only have one observation. We are left with a sample of about 48 thousand tracts.